11 Probability

LESSON SEQUENCE

11.1 Overview 734
11.2 Review of probability 737
11.3 Tree diagrams 754
11.4 Independent and dependent events 762
11.5 Conditional probability 768
11.6 Counting principles and factorial notation (10A) 774
11.7 Review 790



LESSON
11.1 Overview

Why learn this?



Probability is a broad and interesting area of mathematics that affects our day-to-day lives far more than we can imagine. Here is a fun fact: did you know there are so many possible arrangements of the 52 cards in a deck $(52! = 8.0658 \times 10^{67})$ that the probability of ever getting the same arrangement after shuffling is virtually zero? This means every time you shuffle a deck of cards, you are almost certainly producing an arrangement that has never been seen before. Probability is also a big part of computer and board games; for example, the letters X and Q in Scrabble are worth more points because you are less likely to be able to form a word using those letters. It goes without saying that probability is a big part of any casino game and of the odds and payouts when gambling on the outcome of racing or sports.

Although it is handy to know probability factoids and understand gambling, this isn't the reason we spend time learning probability. Probability helps us build critical thinking skills, which are required for success in almost any career and even just for navigating our own lives. For example, if you were told your chance of catching a rare disease had doubled, you probably wouldn't need to worry, as a 1-in-a-million-chance becoming a 2-in-a-million chance isn't a significant increase in the probability of you developing the disease. On the other hand, if a disease has a 1% mortality rate, that may seem fairly low, but it means that if a billion people developed that disease, then 10 million would die. Using probability to understand risk helps us steer clear of manipulation by advertising, politicians and the media. Building on this understanding helps us as individuals to make wise decisions in our day-to-day lives, whether it be investing in the stock market, avoiding habits that increase our risk of sickness, or building our career.

Hey students! Bring these pages to life online

Watch videos

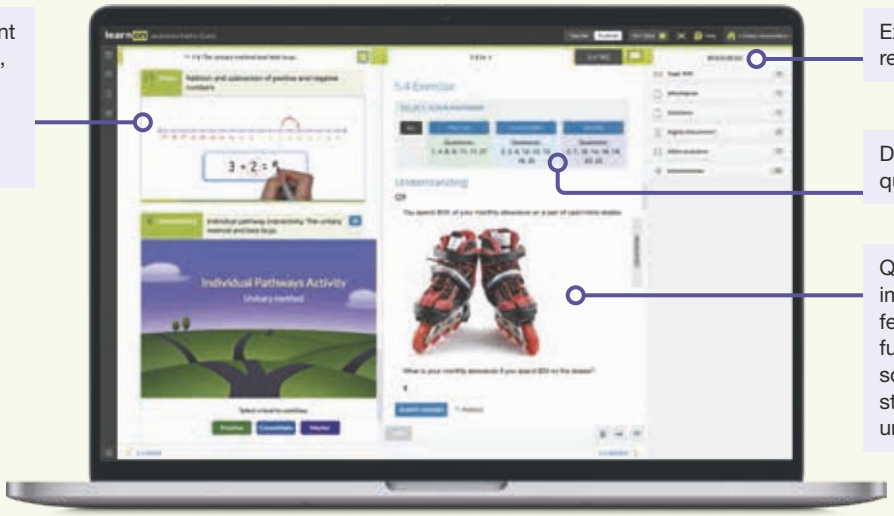
Engage with interactivities

Answer questions and check solutions

Find all this and MORE in jacPLUS



Reading content and rich media, including interactivities and videos for every concept



Extra learning resources

Differentiated question sets

Questions with immediate feedback, and fully worked solutions to help students get unstuck

Exercise 11.1 Pre-test

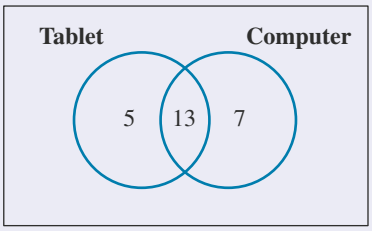
learnon

1. Calculate $\Pr(A \cap B)$ if $\Pr(A = 0.4)$, $\Pr(B = 0.3)$, and $\Pr(A \cup B) = 0.5$

2. If events A and B are mutually exclusive, and $\Pr(B) = 0.38$ and $\Pr(A \cup B) = 0.89$, calculate $\Pr(A)$.

3. State whether the events $A = \{\text{drawing a red marble from a bag}\}$ and $B = \{\text{rolling a 1 on a die}\}$ are independent or dependent.

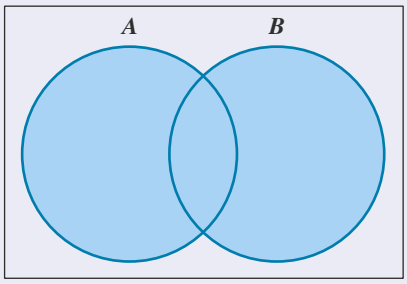
4. The Venn diagram shows the number of university students in a group of 25 who own a computer and/or tablet.
In simplest form, calculate the probability that a university student selected at random will own only a tablet.

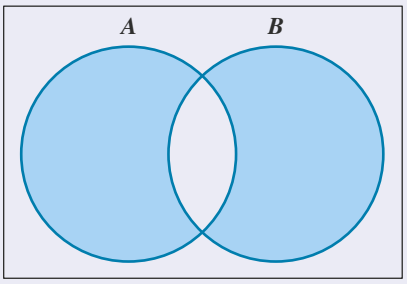


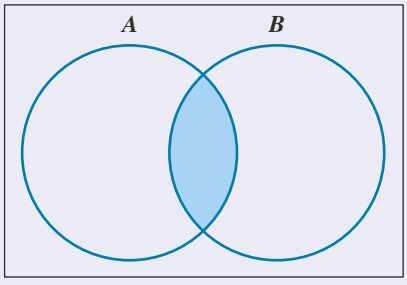
5. MC Two unbiased four-sided dice are rolled. Determine the probability that the total sum of two face-down numbers obtained is 6.

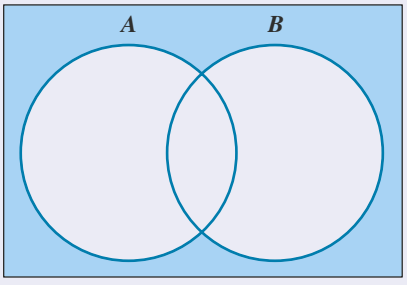
A. $\frac{1}{8}$ B. $\frac{1}{16}$ C. $\frac{3}{16}$ D. $\frac{1}{4}$

6. MC Identify which Venn diagram best illustrates $\Pr(A \cup B)'$.

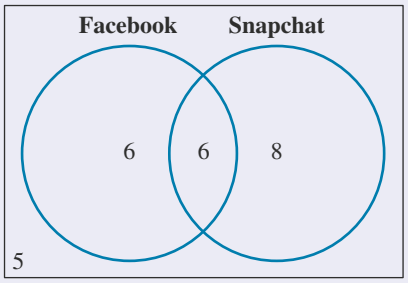
A.


B.


C.


D.


7. MC From a group of 25 people, 12 use Facebook (F), 14 use Snapchat (S), and 6 use both Facebook and Snapchat applications. Determine the probability that a person selected will use neither application.

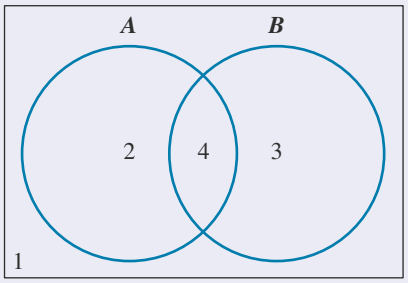


A. $\frac{1}{5}$ B. $\frac{6}{25}$

C. $\frac{8}{25}$ D. $\frac{12}{25}$

8. The probability that a student will catch a bus to school is 0.7 and the independent probability that a student will be late to school is 0.2.
Determine the probability, in simplest form, that a student catches a bus and is not late to school.

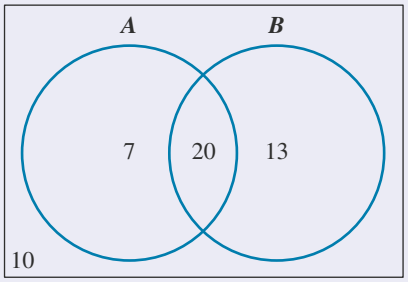
9. MC From events A and B in the Venn diagram, calculate $\Pr(A|B)$.



A. $\frac{4}{7}$ B. $\frac{4}{9}$

C. $\frac{2}{5}$ D. $\frac{1}{3}$

10. In the Venn diagram, events A and B are independent.
State whether this statement is true or false.



11. If $\Pr(A) = 0.5$, $\Pr(B) = 0.4$ and $\Pr(A \cup B) = 0.8$, calculate $\Pr(B|A)$, correct to 1 decimal place.

12. MC From 20 students, 10 play soccer, while 15 play Aussie Rules and 8 play both soccer and Aussie Rules. Calculate the probability that a student randomly selected plays soccer given that he or she plays Aussie Rules.

A. $\frac{8}{15}$ B. $\frac{2}{5}$ C. $\frac{2}{3}$ D. $\frac{10}{23}$

13. MC Two cards are drawn successively without replacement from a pack of playing cards. Calculate the probability of drawing 2 spades.

A. $\frac{1}{17}$ B. $\frac{2}{2652}$ C. $\frac{1}{2652}$ D. $\frac{1}{2704}$

14. A survey of a school of 800 students found that 100 used a bus (B) to get to school, 75 used a train (T) and 650 used neither.
In simplest form, determine the probability that a student uses both a bus and a train to get to school.

15. MC On the first day at school, students are asked to tell the class about their holidays. There are 30 students in the class and all have spent part or all of their holidays at one of the following: a coastal resort, interstate, or overseas.
The teacher finds that:

- 5 students went to a coastal resort only
- 2 students went interstate only
- 2 students holidayed in all three ways
- 8 students went to a coastal resort and travelled overseas only
- 20 students went to a coastal resort
- no less than 4 students went overseas only
- no less than 13 students travelled interstate.

Determine the probability that a student travelled overseas and interstate only.

A. $\frac{2}{15}$ **B.** $\frac{5}{6}$ **C.** $\frac{3}{15}$ **D.** $\frac{1}{3}$

LESSON
11.2 Review of probability

LEARNING INTENTIONS

At the end of this lesson you should be able to:

- use key probability terminology such as trials, frequency, sample space, and likely and unlikely events
- use two-way tables to represent sample spaces
- calculate experimental and theoretical probabilities
- determine the probability of complementary events and mutually exclusive events
- use the addition rule to calculate the probability of event '*A* or *B*'.

eles-4922

11.2.1 The language of probability

- **Probability** measures the chance of an event taking place and ranges from 0 for an impossible event to 1 for a certain event.



- The **experimental probability** of an event is based on the outcomes of experiments, simulations or surveys.
- A **trial** is a single experiment, for example, a single flip of a coin.

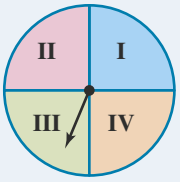
Experimental probability

$$\text{Experimental probability} = \frac{\text{number of successful trials}}{\text{total number of trials}}$$

- The experimental probability of an event is also known as the **relative frequency**.
- The list of all possible outcomes of an experiment is known as the **event space** or **sample space**. For example, when flipping a coin there are two possible outcomes: Heads or Tails. The event space can be written using set notation as {H, T} .

WORKED EXAMPLE 1 Sample space and calculating relative frequency

The spinner shown here is made up of 4 equal-sized segments. It is known that the probability that the spinner will land on any one of the four segments from one spin is $\frac{1}{4}$. To test if the spinner shown here is fair, a student spun the spinner 20 times and each time recorded the segment in which the spinner stopped. The spinner landed as follows.



Segment	I	II	III	IV
Tally	5	4	8	3

a. List the sample space.
b. Given the experimental results, determine the experimental probability of each segment.
c. Compare the experimental probabilities with the known probabilities and suggest how the experiment could be changed to ensure that the results give a better estimate of the true probability.

THINK

a. The sample space lists all possible outcomes from one spin of the spinner. There are four possible outcomes.

b. 1. For segment I there were 5 successful trials out of the 20. Substitute these values into the experimental probability formula.

2. Repeat for segments:
- II (4 successes)
- III (8 successes)
- IV (3 successes).

WRITE

a. Sample space = {I, II, III, IV}

b. $\text{Experimental probability}_{\text{I}} = \frac{\text{number of successful trials}}{\text{total number of trials}}$

$= \frac{5}{20}$

$= 0.25$

$\text{Experimental probability}_{\text{II}} = \frac{4}{20}$

$= 0.2$

$\text{Experimental probability}_{\text{III}} = \frac{8}{20}$

$= 0.4$

$\text{Experimental probability}_{\text{IV}} = \frac{3}{20}$

$= 0.15$

c. Compare the experimental frequency values with the known value of $\frac{1}{4}$ (0.25). Answer the question.	**c.** The experimental probability of segment I was the only segment that mirrored the known value. To ensure that experimental probability gives a better estimate of the true probability, the spinner should be spun many more times.

Simulations

- Sometimes it is not possible to conduct trials of a real experiment because it is too expensive, too difficult or impractical.
- Instead, outcomes of events can be modelled using spinners, dice, coins, or digital tools to simulate what happens in real life.
- These simulated experiments are called **simulations** and can be used to model real-life situations such as the number of people likely to be infected with a virus like COVID-19, or for decision-making such as insurance risk or supply and demand of products.

WORKED EXAMPLE 2 Applying simulation to model real-life situations

Six runners are to compete in a race.
a. Explain how simulation could be used to determine the winner.
b. State any limitations in your answer.



THINK	WRITE
a. There are six runners, so six numbers are needed to represent them. A die is the obvious choice. Assign numbers 1 to 6 to each runner.	Each runner could be assigned a number on a die, which could be rolled to determine the winner of the race. When the die is rolled, it simulates a race. The number that lands uppermost is the winner of the race.
b. In this simulation, each runner had an equally likely chance of winning each game.	The answer depends on each runner having an equally likely chance of winning each game. This is rarely the case.

Two-way tables

- A sample space can be displayed using a **two-way table**.
- A two-way table represents two of the outcomes of events in a two-dimensional table.
  A two-way table for the experiment of tossing a coin and rolling a die simultaneously is shown below.

		Die outcome					
		1	**2**	**3**	**4**	**5**	**6**
Coin outcome	**H**	H, 1	H, 2	H, 3	H, 4	H, 5	H, 6
	T	T, 1	T, 2	T, 3	T, 4	T, 5	T, 6

WORKED EXAMPLE 3 Representing a sample space with a two-way table

Two dice are rolled and the values on the two uppermost faces are multiplied together. Draw a diagram to illustrate the sample space.

THINK

The sample space for rolling 1 die is {1, 2, 3, 4, 5, 6}. When two dice are rolled and the two uppermost faces are multiplied, the sample space is made up of 36 products. This is best represented with the use of a two-way table.

- Draw a 7×7 grid.
- In the first row and column, list the outcomes of each die.
- At the intersection of a column and row, write the product of the relevant die outcomes.

WRITE

	First die						
Second die	×	1	2	3	4	5	6
	1	1	2	3	4	5	6
	2	2	4	6	8	10	12
	3	3	6	9	12	15	18
	4	4	8	12	16	20	24
	5	5	10	15	20	25	30
	6	6	12	18	24	30	36

Theoretical probability

- **Theoretical probability** is the probability of an event occurring, based on the number of possible favourable outcomes, $n(E)$, and the total number of possible outcomes, $n(\xi)$.

Theoretical probability

When all outcomes are equally likely, the theoretical probability of an event can be calculated using the formula:

$$\text{Pr (event)} = \frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}} \quad \text{or} \quad \text{Pr (event)} = \frac{n(E)}{n(\xi)}$$

where $n(E)$ is the number of favourable events and $n(\xi)$ is the total number of possible outcomes.

WORKED EXAMPLE 4 Calculating theoretical probability

A fair die is rolled and the value of the uppermost side is recorded. Calculate the theoretical probability that a 4 is uppermost.

THINK

1. Write the number of favourable outcomes and the total number of possible outcomes. The number of 4s on a fair die is 1. There are 6 possible outcomes.
2. Substitute the values found in part 1 to calculate the probability of the event that a 4 is uppermost when a die is rolled.
3. Write the answer in a sentence.

WRITE

$n(E) = 1$
$n(\xi) = 6$

$$\text{Pr(a 4)} = \frac{n(E)}{n(\xi)} = \frac{1}{6}$$

The probability that a 4 is uppermost when a fair die is rolled is $\frac{1}{6}$.

eles-6142

11.2.2 Properties of probability events

Complementary events

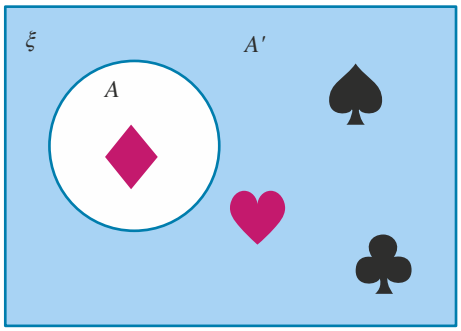
- The **complement** of the set A is the set of all elements that belong to the universal set (ξ) but that do *not* belong to A.
- The complement of A is written as A' and is read as 'A dashed', 'A prime' or 'not A'.
- On a Venn diagram, **complementary events** appear as separate regions that together occupy the whole universal set.

Complementary events

Since complementary events fill the entire sample space:

$$\Pr(A) + \Pr(A') = 1$$

- As an example, the complement of {drawing a diamond} from a deck of cards is {not drawing a diamond}, which can also be described as {drawing a heart, spade or club}.
  This is shown in the Venn diagram.



WORKED EXAMPLE 5 Determining complementary events

A player is chosen from a cricket team. Are the events 'selecting a batter' and 'selecting a bowler' complementary events if a player can have more than one role?
Give a reason for your answer.



THINK

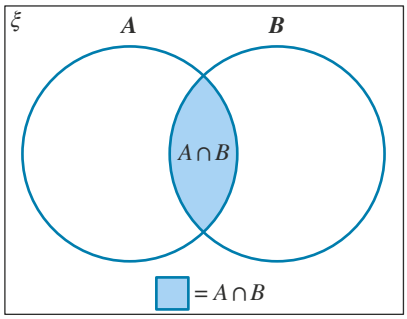
Explain the composition of a cricket team. Players who can bat or bowl are not necessarily the only players in a cricket team. There is a wicket-keeper as well. Some players can both bat and bowl.

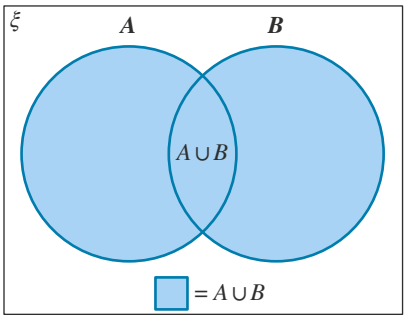
WRITE

No, the events 'selecting a batter' and 'selecting a bowler' are not complementary events. These events may have common elements, that is the players in the team who can bat and bowl. The cricket team also includes a wicket-keeper.

The intersection and union of A and B

- The intersection of two events A and B is written $A \cap B$. These are the outcomes that are in A 'and' in B, so the intersection is often referred to as the event 'A and B'.
- The union of two events A and B is written $A \cup B$. These are the outcomes that are in A 'or' in B, so the union is often referred to as the event 'A or B'.





- When calculating the probability of $A \cup B$, we cannot simply add the probabilities of A and B, as $A \cap B$ would be counted twice.
- The formula for the probability of $A \cup B$ is therefore given by the following equation, which is known as the **Addition Law of probability**.

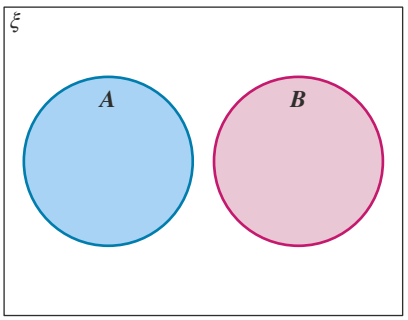
The Addition Law of probability

For intersecting events A and B:

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$$

Mutually exclusive events

- Two events are **mutually exclusive** if one event happening excludes the other from happening. These events may not encompass all possible events.
  For example, when selecting a card from a deck of cards, selecting a black card excludes the possibility that the card is a heart.
- On a Venn diagram, mutually exclusive events appear as disjointed sets within the universal set.
- For mutually exclusive events A and B, $\Pr(A \cap B) = 0$.
  Therefore, the formula for $\Pr(A \cup B)$ is simplified to the following.



Mutually exclusive probabilities

For mutually exclusive events A and B:

$$\Pr(A \cup B) = \Pr(A) + \Pr(B)$$

WORKED EXAMPLE 6 Determining the probability of the union of two events

A card is drawn from a pack of 52 playing cards. Determine the probability that the card is a heart or a club.

THINK	WRITE
1. Determine whether the given events are mutually exclusive.	The two events are mutually exclusive as they have no common elements.
2. Determine the probability of drawing a heart and of drawing a club.	$\Pr(\text{heart}) = \frac{13}{52} = \frac{1}{4}$ $\quad \Pr(\text{club}) = \frac{13}{52} = \frac{1}{4}$
3. Write the Addition Law for two mutually exclusive events.	$\Pr(A \text{ or } B) = \Pr(A) + \Pr(B)$ where A = drawing a heart and B = drawing a club
4. Substitute the known values into the rule.	$\Pr(\text{heart or club}) = \Pr(\text{heart}) + \Pr(\text{club}) = \frac{1}{4} + \frac{1}{4} = \frac{2}{4}$
5. Evaluate and simplify.	$= \frac{1}{2}$
6. Write your answer.	The probability of drawing a heart or a club is $\frac{1}{2}$.
Note: Alternatively, we can use the formula for theoretical probability.	$\Pr(\text{heart or club}) = \frac{n(\text{heart or club})}{n(\xi)} = \frac{26}{52} = \frac{1}{2}$

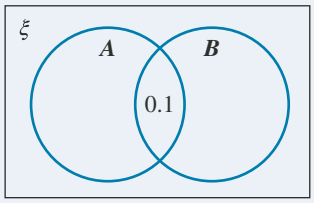
WORKED EXAMPLE 7 Using the Addition Law to determine the intersection of two events

Given $\Pr(A) = 0.6$, $\Pr(B) = 0.4$ and $\Pr(A \cup B) = 0.9$:
a. use the Addition Law of probability to calculate the value of $\Pr(A \cap B)$
b. draw a Venn diagram to represent the universal set
c. calculate $\Pr(A \cap B')$.

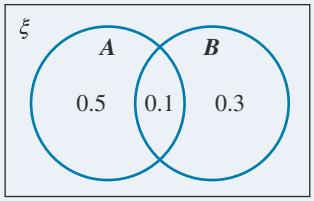
THINK	WRITE
a. 1. Write the Addition Law of probability and substitute given values.	a. $\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$ $0.9 = 0.6 + 0.4 - \Pr(A \cap B)$
2. Collect like terms and rearrange to make $\Pr(A \cap B)$ the subject. Solve the equation.	$0.9 = 1.0 - \Pr(A \cap B)$ $\Pr(A \cap B) = 1.0 - 0.9$ $= 0.1$

b. 1. Draw intersecting sets A and B within the universal set and write $\Pr(A \cap B) = 0.1$ inside the overlapping section, as shown.

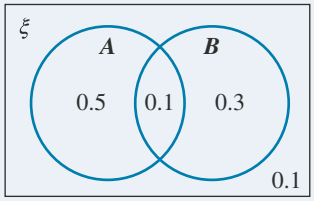
b.



2.
- As $\Pr(A) = 0.6$, 0.1 of this belongs in the overlap; the remainder of set A is 0.5 $(0.6 - 0.1)$.
- Since $\Pr(B) = 0.4$, 0.1 of this belongs in the overlap; the remainder of set B is 0.3 $(0.4 - 0.1)$.

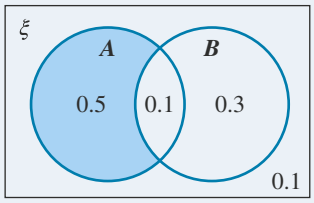


3. The total probability for the universal set is 1. That means $\Pr(A \cup B)' = 0.1$.
Write 0.1 outside sets A and B to form the remainder of the universal set.



c. $\Pr(A \cap B')$ is the overlapping region of $\Pr(A)$ and $\Pr(B')$. Shade the region and write down the corresponding probability value for this area.

c.



$\Pr(A \cap B') = 0.5$

WORKED EXAMPLE 8 Using a Venn diagram to represent sets and find probabilities

a. Draw a Venn diagram representing the relationship between the following sets. Show the position of all the elements in the Venn diagram.
$\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$
$A = \{3, 6, 9, 12, 15, 18\}$
$B = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$

b. Determine:

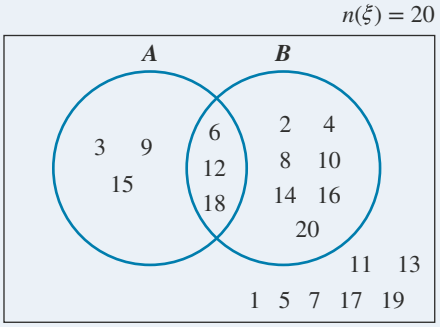
i. **$\Pr(A)$**
ii. **$\Pr(B)$**
iii. **$\Pr(A \cap B)$**
iv. **$\Pr(A \cup B)$**
v. **$\Pr(A' \cap B')$**

THINK

a. 1. Draw a rectangle with two partly intersecting circles labelled A and B.
2. Analyse sets A and B and place any common elements in the central overlap.
3. Place the remaining elements of set A in circle A.
4. Place the remaining elements of set B in circle B.
5. Place the remaining elements of the universal set ξ in the rectangle.

WRITE/DRAW

a.



b. i. 1. Write the number of elements that belong to set A and the total number of elements.

b. i. $n(A) = 6, n(\xi) = 20$

2. Write the rule for probability.

$$\Pr(A) = \frac{n(A)}{n(\xi)}$$

3. Substitute the known values into the rule.

$$\Pr(A) = \frac{6}{20}$$

4. Evaluate and simplify.

$$= \frac{3}{10}$$

ii. 1. Write the number of elements that belong to set B and the total number of elements.

ii. $n(B) = 10, n(\xi) = 20$

2. Repeat steps **2** to **4** of part **b i.**

$$\Pr(B) = \frac{n(B)}{n(\xi)}$$

$$\Pr(B) = \frac{10}{20}$$

$$= \frac{1}{2}$$

iii. 1. Write the number of elements that belong to set $(A \cap B)$ and the total number of elements.

iii. $n(A \cap B) = 3, n(\xi) = 20$

2. Repeat steps **2** to **4** of part **b i.**

$$\Pr(A \cap B) = \frac{n(A \cap B)}{n(\xi)}$$

$$\Pr(A \cap B) = \frac{3}{20}$$

iv. 1. Write the number of elements that belong to set $(A \cup B)$ and the total number of elements.

iv. $n(A \cup B) = 13, n(\xi) = 20$

2. Repeat steps 2 to 4 of part **b i.**

$$n(A \cup B) = \frac{n(A \cup B)}{n(\xi)}$$

$$n(A \cup B) = \frac{13}{20}$$

v. 1. Write the number of elements that belong to set $(A' \cap B')$ and the total number of elements.

v. $n(A' \cap B') = 7, n(\xi) = 20$

2. Repeat steps 2 to 4 of part **b i.**

$$\Pr(A' \cap B') = \frac{n(A' \cap B')}{n(\xi)}$$

$$\Pr(A' \cap B') = \frac{7}{20}$$

WORKED EXAMPLE 9 Creating a Venn diagram for three intersecting events

In a class of 35 students, 6 students like all three subjects: PE, Science and Music. Eight of the students like PE and Science, 10 students like PE and Music, and 12 students like Science and Music. Also, 22 students like PE, 18 students like Science and 17 like Music. Two students don't like any of the subjects.

a. Display this information on a Venn diagram.

b. Determine the probability of selecting a student who:

i. likes PE only

ii. does not like Music.

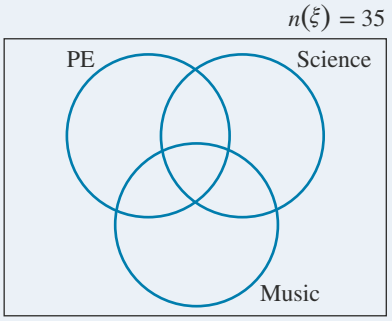
c. Determine Pr[(Science ∪ Music) ∩ PE′].

THINK

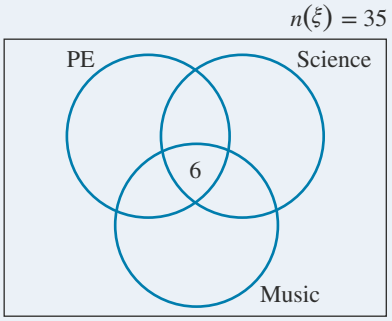
a. 1. Draw a rectangle with three partly intersecting circles, labelled PE, Science and Music.

WRITE/DRAW

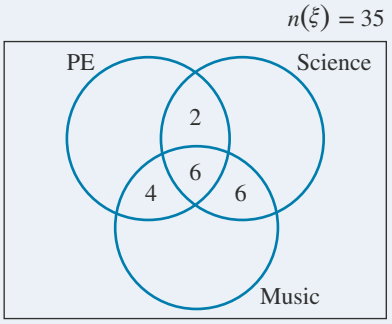
a.



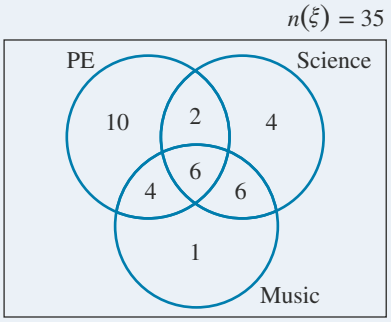
2. Extract the information relating to students liking all three subjects.
Note: The central overlap is the key to solving these problems. Six students like all three subjects, so place the number 6 into the section corresponding to the intersection of the three circles.



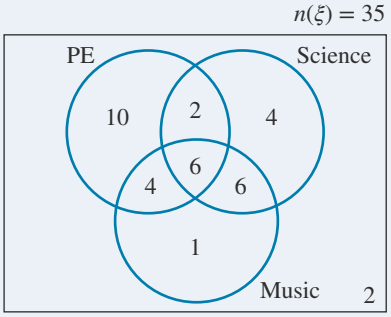
3. Extract the relevant information from the second sentence and place it into the appropriate position.
Note: Eight students like PE and Science; however, 6 of these students have already been accounted for in step 2.
Therefore, 2 will fill the intersection of only PE and Science. Similarly, 4 of the 10 who like and Music will fill the intersection of only PE and Music, and 6 of the 12 students will fill the intersection of only Science and Music.



4. Extract the relevant information from the third sentence and place it into the appropriate position.
Note: Twenty-two students like PE and 12 have already been accounted for in the set. Therefore, 10 students are needed to fill the circle corresponding to PE only. Similarly, 4 students are needed to fill the circle corresponding to Science only to make a total of 18 for Science. One student is needed to fill the circle corresponding to Music only to make a total of 17 for Music.



5. Extract the relevant information from the final sentence and place it into the appropriate position.
Note: Two students do not like any of the subjects, so they are placed in the rectangle outside the three circles.



6. Check that the total number in all positions is equal to the number in the universal set.

$10 + 2 + 4 + 4 + 6 + 6 + 1 + 2 = 35$

b. i. 1. Write the number of students who like PE only and the total number of students in the class.

b. i. $n(\text{students who like PE only}) = 10$
$n(\xi) = 35$

2. Write the rule for probability.

$$\Pr(\text{likes PE only}) = \frac{n(\text{likes PE only})}{n(\xi)}$$

3. Substitute the known values into the rule.

$$\Pr(\text{likes PE only}) = \frac{10}{35}$$

4. Evaluate and simplify.

$$= \frac{2}{7}$$

5. Write your answer.

The probability of selecting a student who likes PE only is $\frac{2}{7}$.

ii. 1. Write the number of students who do not like Music and the total number of students in the class.
Note: Add all the values that do not appear in the Music circle as well as the two that sit in the rectangle outside the circles.

ii. $n(\text{students who do not like Music}) = 18$
$n(\xi) = 35$

2. Write the rule for probability.

$$\Pr(\text{does not like Music}) = \frac{n(\text{does not like Music})}{n(\xi)}$$

3. Substitute the known values into the rule.

$$\Pr(\text{does not like Music}) = \frac{18}{35}$$

4. Write your answer.

The probability of selecting a student who does not like Music is $\frac{18}{35}$.

c. 1. Write the number of students who like Science and Music but not PE.
Note: Add the values that appear in the Science and Music circles but do not overlap with the PE circle.

c. $n[(\text{Science} \cup \text{Music}) \cap \text{PE}'] = 11$
$n(\xi) = 35$

2. Repeat steps **2** to **4** of part **b ii.**

$$\Pr[(\text{Science} \cup \text{Music}) \cap \text{PE}'] = \frac{n[(\text{Science} \cup \text{Music}) \cap \text{PE}']}{n(\xi)}$$

$$\Pr[(\text{Science} \cup \text{Music}) \cap \text{PE}'] = \frac{11}{35}.$$

The probability of selecting a student who likes Science or Music but not PE is $\frac{11}{35}$.

COLLABORATIVE TASK: Probability of matching birthdays

Use simulation to calculate the probability that two people were born on the same day of the year, clearly explaining the method you used. Then research the simulation of the birthday problem and report on your findings.

on Resources

eWorkbook Topic 11 Workbook (worksheets, code puzzle and project) (ewbk-2103)
Video eLesson Venn diagrams (eles-1934)
Interactivities Individual pathway interactivity: Review of probability (int-4616)
Experimental probability (int-3825)
Two-way tables (int-6082)
Theoretical probability (int-6081)
Venn diagrams (int-3828)
Addition Law of probability (int-6168)

Exercise 11.2 Review of probability

learnon

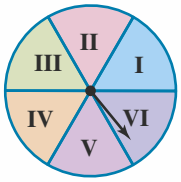
11.2 Quick quiz on | 11.2 Exercise

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 3, 4, 5, 8, 13, 17, 20	2, 6, 9, 10, 14, 18, 21	7, 11, 12, 15, 16, 19, 22, 23

Fluency

1. **WE1** The spinner shown was spun 50 times and the outcome each time was recorded in the table.



Segment	I	II	III	IV	V	VI
Tally	10	6	8	7	12	7

   a. List the event space.
   b. Given the experimental results, determine the relative frequency for each segment.
   c. The theoretical probability of the spinner landing on any particular segment with one spin is $\frac{1}{6}$. State how the experiment could be changed to give a better estimate of the true probabilities.

2. A laptop company conducted a survey to see what were the most appealing colours for laptop computers among 15–18-year-old students. The results were as follows.

Colour	Black Black	Sizzling Silver	Power Pink	Blazing Blue	Gooey Green	Glamour Gold
Number	102	80	52	140	56	70

   a. Calculate the number of students who were surveyed.
   b. Calculate the relative frequency of students who found silver the most appealing laptop colour.
   c. Calculate the relative frequency of students who found black and green to be their most appealing colours.
   d. State which colour was found to be most appealing.

3. **WE2** State the device you would use to simulate the following experiments.
   a. Collecting toys from a particular brand of cereal where there are six in the set
   b. Whether employees are satisfied with their jobs
   c. Whether two people who meet will be born in the same month

4. **WE3** Two dice are rolled and the values on the two uppermost faces are added together.
   a. Construct a table to illustrate the sample space.
   b. Calculate the most likely outcome.
   c. Calculate the least likely outcome.

5. **WE4** A die is rolled. Calculate the probability that the outcome is an even number or a 5.

6. A card is drawn from a well-shuffled pack of 52 playing cards. Calculate:
   a. Pr(a king is drawn)
   b. Pr(a heart is drawn)
   c. Pr(a king or a heart is drawn).

7. WE5 For each of the following pairs of events:

i. state, giving justification, if the pair are complementary events
ii. alter the statements, where applicable, so that the events become complementary events.



a. Having Weet Bix or having Strawberry Pops for breakfast
b. Walking to a friend's place or driving there
c. Watching TV or reading as a leisure activity
d. Rolling a number less than 5 or rolling a number greater than 5 with a ten-sided die with faces numbered 1 to 10
e. Passing a maths test or failing a maths test

8. WE6 A card is drawn from a shuffled pack of 52 cards. Calculate the probability that the card drawn is:

a. an ace
b. a club
c. a red card
d. not a jack
e. a green card
f. not a red card.

9. A bag contains 4 blue marbles, 7 red marbles and 9 yellow marbles. All marbles are the same size. A marble is selected at random. Calculate the probability that the marble is:

a. blue
b. red
c. not yellow
d. black.

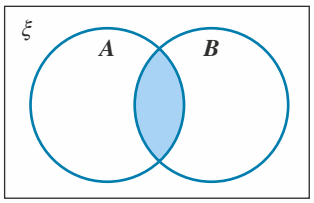
10. WE7 Given $\Pr(A) = 0.5, \Pr(B) = 0.4$ and $\Pr(A \cup B) = 0.8$:

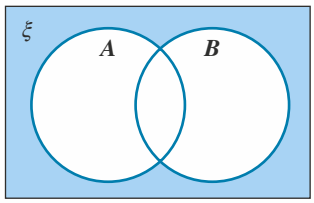
a. use the Addition Law of probability to calculate the value of $\Pr(A \cap B)$
b. draw a Venn diagram to represent the universal set
c. calculate $\Pr(A \cap B')$.

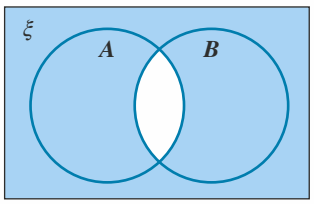
11. Let $\Pr(A) = 0.25, \Pr(B) = 0.65$ and $\Pr(A \cap B) = 0.05$.

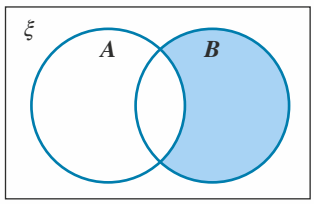
a. Calculate:
i. $\Pr(A \cup B)$
ii. $\Pr(A \cap B)'$.

b. MC Choose which Venn diagram below best illustrates $\Pr(A \cap B)'$.

A.


B.


C.


D.


12. a. WE8 Draw a Venn diagram representing the relationship between the following sets. Show the position of all the elements in the Venn diagram.

$$\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$$
$$A = \{1, 3, 5, 7, 9, 11, 13, 15, 17, 19\}$$
$$B = \{1, 4, 9, 16\}$$

b. Calculate:
i. $\Pr(A)$
ii. $\Pr(B)$
iii. $\Pr(A \cap B)$
iv. $\Pr(A \cup B)$
v. $\Pr(A' \cap B')$.

Understanding

13. You and a friend are playing a dice game. You have an eight-sided die (with faces numbered 1 to 8 inclusive) and your friend has a six-sided die (with faces numbered 1 to 6 inclusive).
You each roll your own die.

- **a.** The person who rolls the number 4 wins. Determine if this game is fair.
- **b.** The person who rolls an odd number wins. Determine if this game is fair.

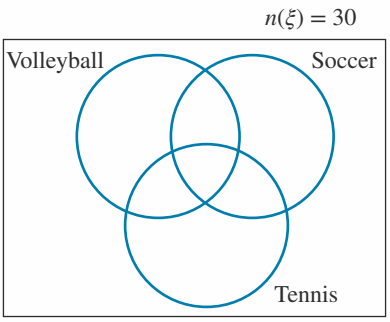
14. A six-sided die has three faces numbered 5; the other faces are numbered 6. Determine if the events 'rolling a 5' and 'rolling a 6' are equally likely.

15. (*Note:* You will need a six-sided die for this question.)
The latest promotion by a chocolate bar manufacturer claims 'Every third one is free'. When you get the special message inside the wrapper, you get a refund on the purchase price.
The manufacturer has put the special message in every third wrapper. When the chocolate bars are packaged into boxes, they are mixed randomly.
Use the numbers 3 and 6 on a die to represent getting a free chocolate bar.

- **a.** Carry out 50 trial purchases of chocolate bars.
- **b.** Calculate the relative frequency of a free chocolate bar.
- **c.** Determine how many free chocolate bars you would expect to receive in 100 trials.

16. WE9 Thirty students were asked which lunchtime sports they enjoyed — volleyball, soccer or tennis.
Five students chose all three sports. Six students chose volleyball and soccer, 7 students chose volleyball and tennis, and 9 chose soccer and tennis.
Fifteen students chose volleyball, 14 students chose soccer and 18 students chose tennis.

- **a.** Copy the Venn diagram shown and enter the given information.
- **b.** If a student is selected at random, determine the probability of selecting a student who:
  - **i.** chose volleyball
  - **ii.** chose all three sports
  - **iii.** chose both volleyball and soccer but not tennis
  - **iv.** did not choose tennis
  - **v.** chose soccer.
- **c.** Determine:
  - **i.** $\Pr[(\text{soccer} \cup \text{tennis}) \cap \text{volleyball}']$
  - **ii.** $\Pr[(\text{volleyball} \cup \text{tennis}) \cap \text{soccer}']$.



Reasoning

17. A six-sided die has three faces numbered 1 and the other three faces numbered 2. Determine if the events 'rolling a 1' and 'rolling a 2' are equally likely.

18. With the use of diagrams, show that $\Pr(A' \cap B') = \Pr(A \cup B)'$.

19. A drawer contains purple socks and red socks. The chance of obtaining a red sock is 2 in 9. There are 10 red socks in the drawer.
Determine the smallest number of socks that need to be added to the drawer so that the probability of drawing a red sock increases to 3 in 7.

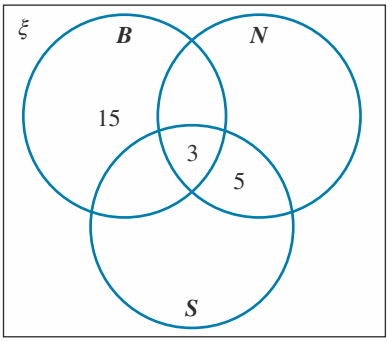


Problem solving

20. Ninety students were asked which lunchtime sports on offer, basketball, netball and soccer, they had participated in on at least one occasion in the last week.
The results are shown in the following table.

Sport	Basketball	Netball	Soccer	Basketball and netball	Basketball and soccer	Netball and soccer	All three
Number of students	35	25	39	5	18	8	3

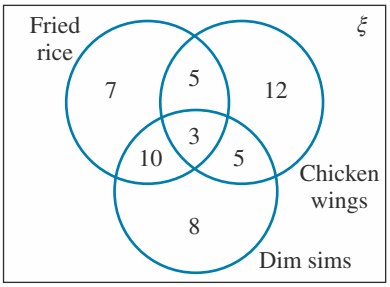
a. Copy and complete the Venn diagram shown below to illustrate the sample space.





b. Determine how many students did not play basketball, netball or soccer at lunchtime.
c. Determine how many students played basketball and/or netball but not soccer.
d. Determine how many students are represented by the region (basketball ∩ not netball ∩ soccer).
e. Calculate the relative frequency of the region described in part **d** above.
f. Estimate the probability that a student will play three of the sports offered.

21. The Venn diagram shows the results of a survey completed by a Chinese restaurateur to find out the food preferences of his regular customers.



a. Determine the number of customers:
- **i.** surveyed
- **ii.** showing a preference for fried rice only
- **iii.** showing a preference for fried rice
- **iv.** showing a preference for chicken wings and dim sims.

b. A customer from this group won the draw for a lucky door prize. Determine the probability that this customer:
- **i.** likes fried rice
- **ii.** likes all three — fried rice, chicken wings and dim sims
- **iii.** prefers chicken wings only.

c. A similar survey was conducted a month later with another group of 50 customers. This survey yielded the following results:
2 customers liked all three foods; 6 preferred fried rice and chicken wings only;
7 preferred chicken wings and dim sims only; 8 preferred fried rice and dim sims only; 22 preferred fried rice; 23 preferred chicken wings; and 24 preferred dim sims.
- **i.** Display this information on a Venn diagram.
- **ii.** Determine the probability of selecting a customer who prefers all three foods, if a random selection is made.

22. A pair of dice is rolled and the sum of the numbers shown is noted.

a. Show the sample space in a two-way table.

b. Determine how many different ways the sum of 7 can be obtained.

c. Determine if all outcomes are equally likely.

d. Complete the given table.

Sum	2	3	4	5	6	7	8	9	11	12
Frequency										

e. Determine the relative frequencies of the following sums.

i. 2 **ii.** 7 **iii.** 11

f. Determine the probability of obtaining the following sums.

i. 2 **ii.** 7 **iii.** 11

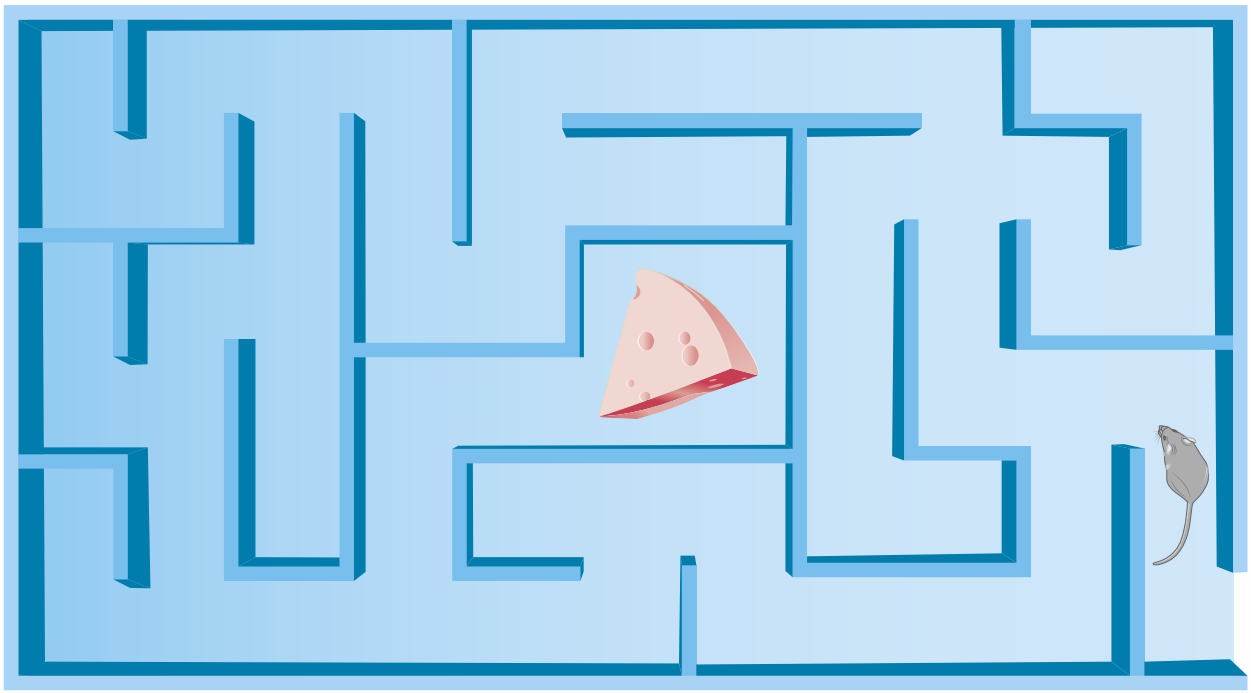
g. If a pair of dice is rolled 300 times, calculate how many times you would expect the sum of 7.

23. The mouse in the maze is trying to reach the cheese. Use a coin to help the mouse to decide to go left or right when a new choice is available. If you come to a dead end, turn around and go back to where you have a choice to go left or right.

There are two possible outcomes for the mouse:

- escaping the maze
- reaching the cheese.

Use simulation to determine the relative frequencies of each of the outcomes.



LESSON
11.3 Tree diagrams

LEARNING INTENTIONS

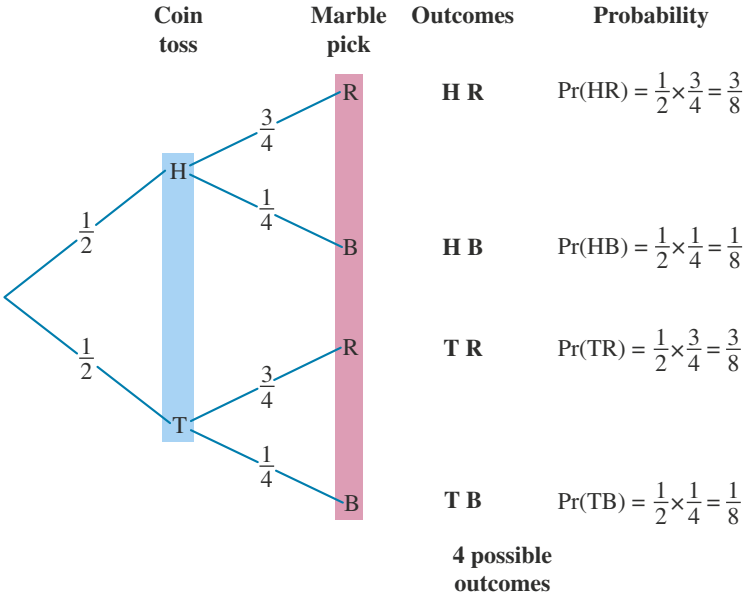
At the end of this lesson you should be able to:

- create tree diagrams to represent two- and three-step chance experiments
- use tree diagrams to solve probability problems involving two or more trials or events.

11.3.1 Two-step chance experiments

eles-4924

- In **two-step chance experiments** the result is obtained after performing two trials. Two-step chance experiments are often represented using tree diagrams.
- **Tree diagrams** are used to list all possible outcomes of two or more events that are not necessarily equally likely.
- The probability of obtaining the result for a particular event is listed on the branches.
- The probability for each outcome in the sample space is the product of the probabilities associated with the respective branches. For example, the tree diagram shown here represents the sample space for flipping a coin and then choosing a marble from a bag containing three red marbles and one black marble.



- When added together, all the probabilities for the outcomes should sum to 1. They are complementary events.
  For example,

$$\begin{aligned}\Pr(HR) + \Pr(HB) + \Pr(TR) + \Pr(TB) &= \frac{3}{8} + \frac{1}{8} + \frac{3}{8} + \frac{1}{8} \\ &= 1\end{aligned}$$

- Other probabilities can also be calculated from the tree diagram.
  For example, the probability of getting an outcome that contains a red marble can be calculated by summing the probabilities of each of the possible outcomes that include a red marble.

Outcomes that contain a red marble are HR and TR. Therefore:

$$\begin{aligned}\Pr(\text{red marble}) &= \Pr(\text{HR}) + \Pr(\text{TR}) \\ &= \frac{3}{8} + \frac{3}{8} \\ &= \frac{6}{8} \\ &= \frac{3}{4}\end{aligned}$$

WORKED EXAMPLE 10 Using a tree diagram for a two-step chance experiment

A three-sided die is rolled and an object is picked out of a hat that contains 3 balls and 7 cubes.

a. Construct a tree diagram to display the sample space.

b. Calculate the probability of:

i. rolling a 3, then choosing a cube

ii. choosing a cube after rolling an odd number.

THINK

a. 1. Draw 3 branches from the starting point to show the 3 possible outcomes of rolling a three-sided die (shown in blue) and then draw 2 branches off each of these to show the 2 possible outcomes of choosing an object (shown in red).

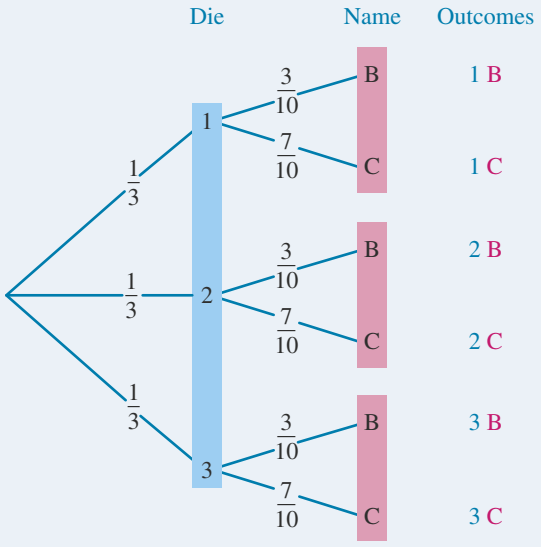
2. Write probabilities on the branches to show the individual probabilities of rolling a 1, 2 or 3 on a three-sided die. As these are equally likely outcomes, $\Pr(1) = \Pr(2) = \Pr(3) = \frac{1}{3}$.

3. Write probabilities on the branches to show the individual probabilities of choosing an object. Since there are 3 balls and 7 cubes in the hat, $\Pr(\text{B}) = \frac{3}{10}$ and $\Pr(\text{C}) = \frac{7}{10}$.

b. i. 1. Follow the pathway of rolling a 3 $\left[\Pr(3) = \frac{1}{3}\right]$ and choosing a cube $\left[\Pr(\text{C}) = \frac{7}{10}\right]$, and multiply the probabilities.

WRITE

a.



b. i. $\Pr(3\text{C}) = \Pr(3) \times \Pr(\text{C})$

$$= \frac{1}{3} \times \frac{7}{10}$$

$$= \frac{7}{30}$$

2. Write the answer.

The probability of rolling a 3 and then choosing a cube is $\frac{7}{30}$.

ii. 1. To roll an odd number (1 or 3) and then choose a cube:
- roll a 1 and then choose a cube, or
- roll a 3 and then choose a cube.

Calculate the probability of each of these and add them together to calculate the total probability. Simplify the result if possible.

ii. $\Pr(\text{odd C}) = \Pr(1\text{C}) + \Pr(3\text{C})$

$= \Pr(1) \times \Pr(\text{C}) + \Pr(3) \times \Pr(\text{C})$

$= \frac{1}{3} \times \frac{7}{10} + \frac{1}{3} \times \frac{7}{10}$

$= \frac{7}{30} + \frac{7}{30}$

$= \frac{14}{30}$

$= \frac{7}{15}$

2. Write the answer.

The probability of choosing a cube after rolling an odd number is $\frac{7}{15}$.

eles-4925

11.3.2 Three-step chance experiments

- In **three-step chance experiments**, the result is obtained after performing three trials. Three-step chance experiments can also be represented using tree diagrams.
- Outcomes are often made up of combinations of events. For example, when a coin is flipped three times, three of the possible outcomes are HHT, HTH and THH. These outcomes all contain 2 Heads and 1 Tail.
- The probability of an outcome with a particular order is written such that the order required is shown. For example, Pr(HHT) is the probability of H on the first coin, H on the second coin and T on the third coin.
- The probability of an outcome with a particular combination of events in which the order is not important is written describing the particular combination required, for example Pr(2 Heads and 1 Tail).



WORKED EXAMPLE 11 Using a tree diagram for a three-step chance experiment

A coin is biased so that the chance of it falling as a Head when flipped is 0.75.

a. Construct a tree diagram to represent the coin being flipped three times.

b. Calculate the following probabilities, correct to 3 decimal places:

i. Pr(HTT)

ii. Pr(1H and 2T)

iii. Pr(at least 2 Tails).

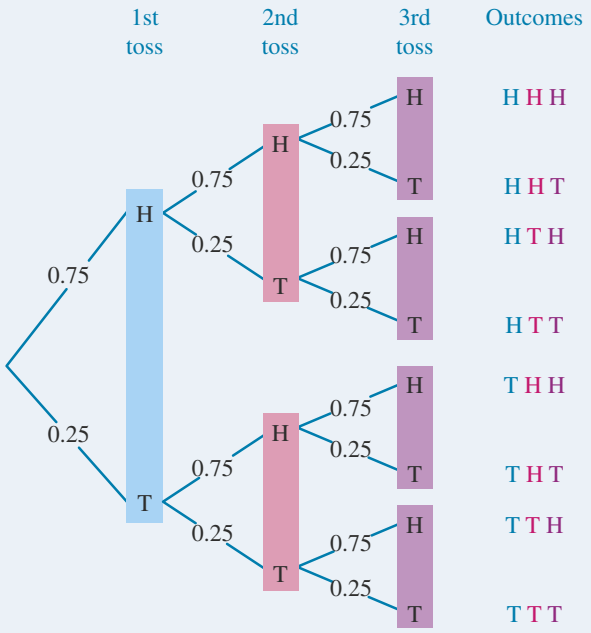
THINK

a. 1. Tossing a coin has two outcomes. Draw 2 branches from the starting point to show the first toss, 2 branches off each of these to show the second toss, and 2 branches off each of these to show the third toss.

2. Write probabilities on the branches to show the individual probabilities of tossing a Head (0.75) and a Tail. Because tossing a Head and tossing a Tail are mutually exclusive, $\Pr(T) = 1 - \Pr(H) = 0.25$.

WRITE

a.



b. i. 1. Pr(HTT) implies the order H(0.75), T (0.25) and T (0.25).

2. Multiply the probabilities and round.

b. i. $\Pr(HTT) = \Pr(H) \times \Pr(T) \times \Pr(T)$
$= (0.75) \times (0.25)^2$
$= 0.046875$
≈ 0.047

ii. 1. Pr(1H and 2T) implies Pr(HTT), Pr(THT) and Pr(TTH).

2. Multiply the probabilities and round.

ii. $\Pr(1H \text{ and } 2T) = \Pr(HTT) + \Pr(THT) + \Pr(TTH)$
$= 3(0.75 \times 0.25^2)$
$= 0.140625$
≈ 0.141

iii. 1. Pr(at least 2 Tails) implies Pr(HTT), Pr(THT), Pr(TTH) and Pr(TTT).

2. Add these probabilities and round.

iii. $\Pr(\text{at least } 2T) = \Pr(HTT) + \Pr(THT) + \Pr(TTH) + \Pr(TTT)$
$= 3(0.75 \times 0.25^2) + 0.25^3$
$= 0.15625$
≈ 0.156

TI | THINK

a. i.

In a new document on a Calculator page, to calculate the probability of an exact number of successes from a set number of trials. press:

- MENU
- 5: Probability
- 5: Distributions
- A: BinomialPdf

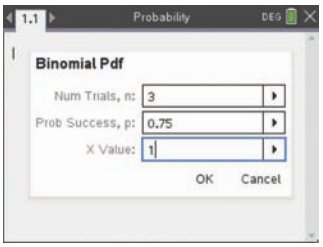
Enter the values:

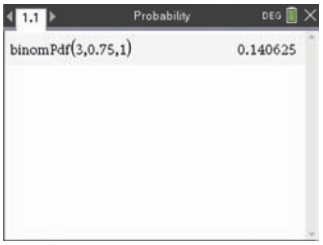
- Num Trials, n: 3
- Prob Success, p: 0.75
- X Value: 1

Then tap OK.

DISPLAY/WRITE

a. i.





Pr(1H and 2T) = 0.141 (correct to 3 decimal places)

ii.

In a new document on a Calculator page, to calculate the probability of at least x successes from a set number of trials, press:

- MENU
- 5: Probability
- 5: Distributions
- B: BinomialCdf

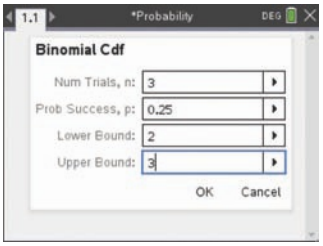
Enter the values:

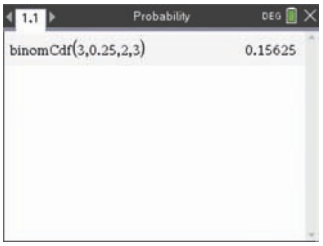
- Num Trials, n: 3
- Prob Success, p: 0.25
- Lower Bound: 2
- Upper Bound: 3

Then tap OK.

Note: In situations with repeated trials of independent events, such as flipping a coin, it becomes unfeasible to draw a tree diagram when the number of trials is 4 or more. BinomialPdf and BinomialCdf are very helpful in these cases.

ii.





Pr(at least 2T) = 0.156 (correct to 3 decimal places)

CASIO | THINK

a. i.

On the Main screen, to calculate the probability of an exact number of successes from a set number of trials, tap:

- Interactive
- Distribution/Inv. Dist
- Discrete
- binomialPDF

Enter the values:

- X: 1
- Numtrial: 3
- pos: 0.75

Then tap OK.

DISPLAY/WRITE

a. i.





Pr(1H and 2T) = 0.141 (correct to 3 decimal places)

ii.

On the Main screen, to calculate the probability of at least x successes from a set number of trials, tap:

- Interactive
- Distribution/Inv. Dist
- Discrete
- binomialCDF

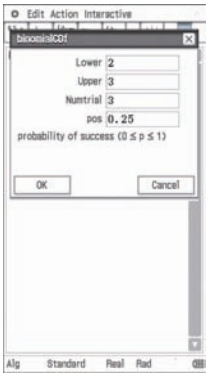
Enter the values:

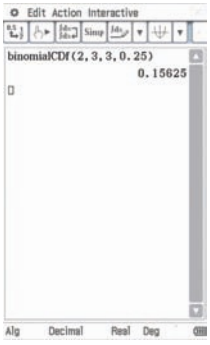
- Lower: 2
- Upper: 3
- Numtrial: 3
- pos: 0.25

Then tap OK.

Note: In situations with repeated trials of independent events, such as flipping a coin, it becomes unfeasible to draw a tree diagram when the number of trials is 4 or more. BinomialPDF and BinomialCDF are very helpful in these cases.

ii.





Pr(at least 2T) = 0.156 (correct to 3 decimal places)

Resources

eWorkbook	Topic 11 Workbook (worksheets, code puzzle and project) (ewbk-2103)
Video eLesson	Tree diagrams (eles-1894)
Interactivities	Individual pathway interactivity: Tree diagrams (int-4617) Tree diagrams (int-6171)

Exercise 11.3 Tree diagrams

learnon

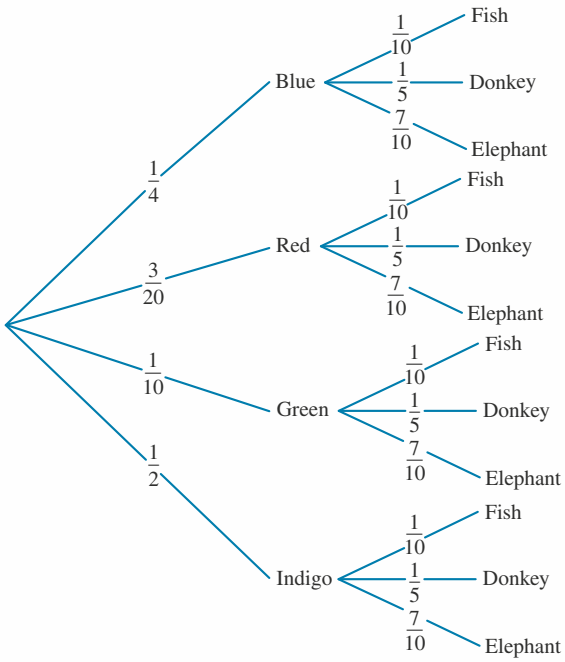
11.3 Quick quiz on | 11.3 Exercise

Individual pathways

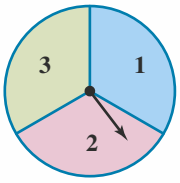
PRACTISE	CONSOLIDATE	MASTER
1, 2, 6, 9, 12	3, 5, 7, 10, 13	4, 8, 11, 14

Fluency

1. Explain how a tree diagram can be used to calculate probabilities of events that are not equally likely.
2. Use the tree diagram shown to answer the following questions.
   a. Identify how many different outcomes there are.
   b. Explain whether all outcomes are equally likely.
   c. State whether getting a red fish is more, less or equally likely than getting a green elephant.
   d. Determine the most likely outcome.
   e. Calculate the following probabilities.
      i. Pr(blue elephant)
      ii. Pr(indigo elephant)
      iii. Pr(donkey)



3. **WE10** The spinner shown is divided into 3 equal-sized wedges labelled 1, 2 and 3. The spinner is spun three times, and it is noted whether the spinner lands on a prime number, Pr = {2, 3} = 'prime', or not a prime number, Pr′ = {1} = 'not prime'.
   a. Construct a labelled tree diagram for 3 spins of the spinner, showing probabilities on the branches and all possible outcomes.
   b. Calculate the following probabilities.
      i. Pr(3 prime numbers)
      ii. Pr(PPP′ in this order)
      iii. Pr(PPP′ in any order)



4. This question relates to rolling a fair die.

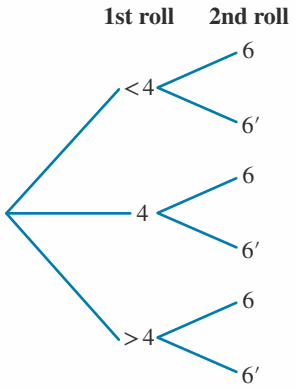
   a. Calculate the probability of each of the following outcomes from one roll of a die.

      i. Pr(rolling number < 4)
      ii. Pr(rolling a 4)
      iii. Pr(rolling a number other than a 6)

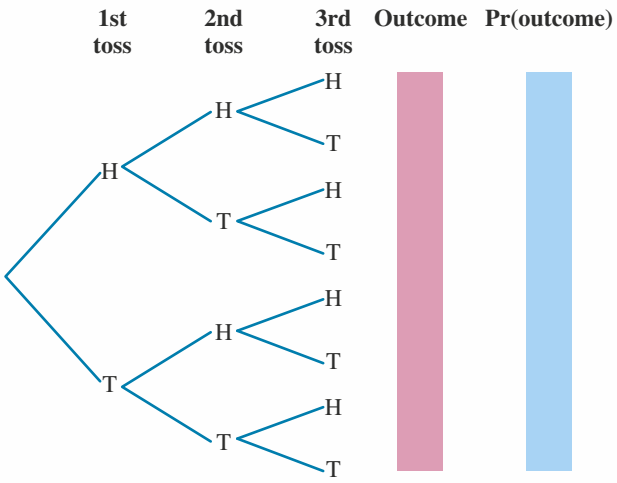
   b. The tree diagram shown has been condensed to depict rolling a die twice, noting the number relative to 4 on the first roll and 6 on the second. Complete a labelled tree diagram, showing probabilities on the branches and all outcomes, similar to that shown.

   c. Determine the probability of rolling the following with 2 rolls of the die.

      i. Pr(a 4 then a 6)
      ii. Pr(a number less than 4 then a 6)
      iii. Pr(a 4 then 6′)
      iv. Pr(a number > 4 then a number < 6)



5. a. Copy the tree diagram shown and complete the labelling for tossing a biased coin three times when the chance of tossing one Head in one toss is 0.7.



   b. Calculate the probability of tossing three Heads.
   c. Determine the probability of getting at least one Tail.
   d. Calculate the probability of getting exactly two Tails.

Understanding

6. **WE11** A coin is biased so that the chance of it falling as a Tail when tossed is 0.2.

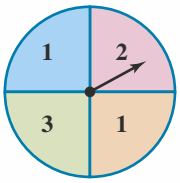
   a. Construct a tree diagram to represent the coin being tossed 3 times.
   b. Determine the probability of getting the same outcome on each toss.

7. A die is tossed twice and each time it is recorded whether or not the number is a multiple of 3. If M = the event of getting a multiple of 3 on any one toss and M' = the event of not getting a multiple of 3 on any one toss:

   a. construct a tree diagram to represent the 2 tosses
   b. calculate the probability of getting two multiples of 3.

8. The biased spinner illustrated is spun three times.

   a. Construct a completely labelled tree diagram for 3 spins of the spinner, showing probabilities on the branches and all possible outcomes and associated probabilities.
   b. Calculate the probability of getting exactly two 1s.
   c. Calculate the probability of getting at most two 1s.



Reasoning

9. Each morning when Ahmed gets dressed for work he has the following choices:

- three suits that are grey, blue and white
- four shirts that are white, blue, pink and grey
- two ties that are grey and blue.

a. Construct a fully labelled tree diagram showing all possible clothing choices.
b. Calculate the probability of picking a grey suit, a pink shirt and a blue tie.
c. Calculate the probability of picking the same colour for all three options.
d. Ahmed works five days a week for 48 weeks of the year. Determine how many times each combination would get repeated over the course of one year at work.
e. Determine how many more combinations of clothing he would have if he bought another tie and another shirt.

10. A restaurant offers its customers a three-course dinner, where they choose between two entrées, three main meals and two desserts. The managers find that 30% choose soup and 70% choose prawn cocktail for the entrée; 20% choose vegetarian, 50% chicken, and the rest have beef for their main meal; and 75% have sticky date pudding while the rest have apple crumble for dessert.



a. Construct a fully labelled tree diagram showing all possible choices.
b. Determine the probability that a customer will choose soup, chicken and sticky date pudding.
c. If there are 210 people booked for the following week at the restaurant, determine how many you would expect to have the meal combination referred to in part **b**.

11. A bag contains 7 red and 3 white balls. A ball is taken at random, its colour is noted, and it is then placed back in the bag before a second ball is chosen at random and its colour is noted.

a. i. Show the possible outcomes with a fully labelled tree diagram.
ii. As the first ball was chosen, determine how many balls were in the bag.
iii. As the second ball was chosen, determine how many balls were in the bag.
iv. Explain whether the probability of choosing a red or white ball changes from the first selection to the second.
v. Calculate the probability of choosing a red ball twice.
b. Suppose that after the first ball had been chosen it was not placed back in the bag.
i. As the second ball is chosen, determine how many balls are in the bag.
ii. Explain if the probability of choosing a red or white ball changes from the first selection to the second.
iii. Construct a fully labelled tree diagram to show all possible outcomes.
iv. Evaluate the probability of choosing two red balls.

Problem solving

12. An eight-sided die is rolled three times to see whether 5 occurs.

a. Construct a tree diagram to show the sample space.
b. Calculate:
i. Pr(three 5s) **ii.** Pr(no 5s) **iii.** Pr(two 5s) **iv.** Pr(at least two 5s).

13. A tetrahedral die (four faces labelled 1, 2, 3 and 4) is rolled and a coin is tossed simultaneously.
    a. Show all the outcomes on a two-way table.
    b. Construct a tree diagram and list all outcomes and their respective probabilities.
    c. Determine the probability of getting a Head on the coin and an even number on the die.
14. A biased coin that has an 80% chance of getting a Head is flipped four times. Use a tree diagram to answer to the following.
    a. Calculate the probability of getting 4 Heads.
    b. Determine the probability of getting 2 Heads then 2 Tails in that order.
    c. Calculate the probability of getting 2 Heads and 2 Tails in any order.
    d. Determine the probability of getting more Tails than Heads.

LESSON
11.4 Independent and dependent events

LEARNING INTENTIONS

At the end of this lesson you should be able to:
- determine whether two events are dependent or independent
- calculate the probability of outcomes from two-step chance experiments involving independent or dependent events.

11.4.1 Independent events

eles-4926

- **Independent events** are events that have no effect on each other. The outcome of the first event does not influence the outcome of the second.
- An example of independent events is successive coin tosses. The outcome of the first toss has no effect on the second coin toss. Similarly, the outcome of the roll on a die will not affect the outcome of the next roll.
- If events A and B are independent, then the **Multiplication Law of probability** states that:
  $\Pr(A \text{ and } B) = \Pr(A) \times \Pr(B)$ or $\Pr(A \cap B) = \Pr(A) \times \Pr(B)$
- The reverse is also true. If:
  $\Pr(A \text{ and } B) = \Pr(A) \times \Pr(B)$ or $\Pr(A \cap B) = \Pr(A) \times \Pr(B)$
  then event A and event B are independent events.

Independent events

For independent events A and B:

$$\Pr(A \cap B) = \Pr(A) \times \Pr(B)$$

WORKED EXAMPLE 12 Calculating probability if two events are independent

If A and B are independent events and $\Pr(A) = 0.6$ and $\Pr(B) = 0.3$, calculate:
a. $\Pr(A \cap B)$ **b.** $\Pr(A' \cap B)$ **c.** $\Pr(A \cap B')$ **d.** $\Pr(A' \cap B')$

THINK

a. The events are independent, so the formula $\Pr(A \cap B) = \Pr(A) \times \Pr(B)$ can be applied.

WRITE

$$\begin{aligned}\Pr(A \cap B) &= \Pr(A) \times \Pr(B) \\ &= 0.6 \times 0.3 \\ &= 0.18\end{aligned}$$

b. **1.** A' is the complement of A, so $\Pr(A')$ can be calculated using $\Pr(A') = 1 - \Pr(A)$.

$$\begin{aligned}\Pr(A') &= 1 - \Pr(A)\\ &= 1 - 0.6\\ &= 0.4\end{aligned}$$

2. Then apply the formula for independent events.

$$\begin{aligned}\Pr(A' \cap B) &= \Pr(A') \times \Pr(B)\\ &= 0.4 \times 0.3\\ &= 0.12\end{aligned}$$

c. B' is the complement of B, so $\Pr(B')$ can be calculated using $\Pr(B') = 1 - \Pr(B)$.

$$\begin{aligned}\Pr(B') &= 1 - \Pr(B)\\ &= 1 - 0.3\\ &= 0.7\end{aligned}$$

$$\begin{aligned}\Pr(A \cap B') &= \Pr(A) \times \Pr(B')\\ &= 0.6 \times 0.7\\ &= 0.42\end{aligned}$$

d. Use the formula for independent events.

$$\begin{aligned}\Pr(A' \cap B') &= \Pr(A') \times \Pr(B')\\ &= 0.4 \times 0.7\\ &= 0.28\end{aligned}$$

11.4.2 Dependent events

eles-4927

- **Dependent events** are events where the outcome of one event affects the outcome of the other event.
- An example of dependent events is drawing a card from a deck of playing cards. The probability that the first card drawn is a heart, Pr(heart), is $\frac{13}{52}$ (or $\frac{1}{4}$).
  If this card is a heart and is not replaced, then this will affect the probability of subsequent draws. The probability that the second card drawn is a heart will be $\frac{12}{51}$, and the probability that the second card is not a heart will be $\frac{39}{51}$.
- If two events are dependent, then the probability of the occurrence of one event affects that of the subsequent event.

WORKED EXAMPLE 13 Calculating the probability of outcomes of dependent events

A bag contains 5 blue, 6 green and 4 yellow marbles. The marbles are identical in all respects except their colours. Two marbles are picked in succession without replacement. Calculate the probability of picking 2 blue marbles.

THINK

1. Determine the probability of picking the first blue marble.

WRITE

$$\Pr(\text{picking a blue marble}) = \frac{n(B)}{n(\xi)}$$

$$\Pr(\text{picking a blue marble}) = \frac{5}{15}$$

$$= \frac{1}{3}$$

2. Determine the probability of picking the second blue marble.
Note: The two events are dependent since marbles are not being replaced. Since we have picked a blue marble, this leaves 4 blue marbles remaining out of a total of 14 marbles.

$$\Pr(\text{picking second blue marble}) = \frac{n(B)}{n(\xi)}$$
$$\Pr(\text{picking second blue marble}) = \frac{4}{14}$$
$$= \frac{2}{7}$$

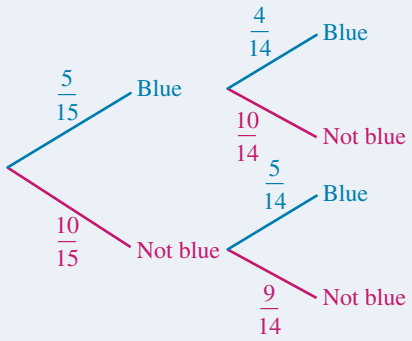
3. Calculate the probability of obtaining 2 blue marbles.

$$\Pr(\text{2 blue marbles}) = \Pr(\text{1st blue}) \times \Pr(\text{2nd blue})$$
$$= \frac{1}{3} \times \frac{2}{7}$$
$$= \frac{2}{21}$$

4. Write your answer.

The probability of obtaining 2 blue marbles is $\frac{2}{21}$.

Note: Alternatively, a tree diagram could be used to solve this question.
The probability of selecting 2 blue marbles successively can be read directly from the first branch of the tree diagram.



$$\Pr(\text{2 blue marbles}) = \frac{5}{15} \times \frac{4}{14}$$
$$= \frac{1}{3} \times \frac{2}{7}$$
$$= \frac{2}{21}$$

on Resources

eWorkbook Topic 11 Workbook (worksheets, code puzzle and project) (ewbk-2103)

Interactivities Individual pathway interactivity: Independent and dependent events (int-4618)
Independent and dependent events (int-2787)
Multiplication Law of probability (int-6172)
Dependent events (int-6173)

Exercise 11.4 Independent and dependent events

learnon

11.4 Quick quiz on | 11.4 Exercise

Individual pathways

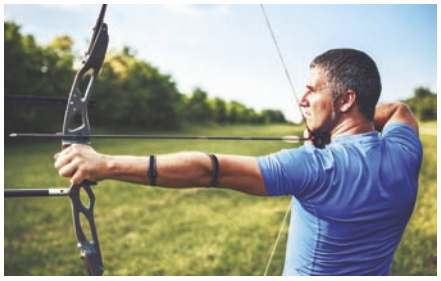
PRACTISE	CONSOLIDATE	MASTER
1, 4, 8, 11, 14, 16, 20	2, 5, 9, 12, 15, 17, 21	3, 6, 7, 10, 13, 18, 19, 22

Fluency

1. WE12 If A and B are independent events and $\Pr(A) = 0.7$ and $\Pr(B) = 0.4$, calculate:
   - a. $\Pr(A \text{ and } B)$
   - b. $\Pr(A' \text{ and } B)$ where A' is the complement of A
   - c. $\Pr(A \text{ and } B')$ where B' is the complement of B
   - d. $\Pr(A' \text{ and } B')$.
2. Determine whether two events A and B with $\Pr(A) = 0.6$, $\Pr(B') = 0.84$ and $\Pr(A \cap B) = 0.96$ are independent or dependent.
3. Determine whether two events A and B with $\Pr(A) = 0.25$, $\Pr(B) = 0.72$ and $\Pr(A \cup B) = 0.79$ are independent or dependent.

Understanding

4. A die is rolled and a coin is tossed.
   - a. Explain whether the outcomes are independent.
   - b. Calculate:
     - i. Pr(Head) on the coin
     - ii. Pr(6) on the die.
   - c. Calculate Pr(6 on the die and Head on the coin).
5. A tetrahedral (4-faced) die and a 10-sided die are rolled simultaneously. Calculate the probability of getting a 3 on the tetrahedral die and an 8 on the 10-sided die.
6. A blue die and a green die are rolled. Calculate the probability of getting a 5 on the blue die and not a 5 on the green die.
7. Dean is an archer. The experimental probability that Dean will hit the target is $\frac{4}{5}$.



   - a. Calculate the probability that Dean will hit the target on two successive attempts.
   - b. Calculate the probability that Dean will hit the target on three successive attempts.
   - c. Calculate the probability that Dean will not hit the target on two successive attempts.
   - d. Calculate the probability that Dean will hit the target on the first attempt but miss on the second attempt.

8. MC A bag contains 20 apples, of which 5 are bruised. Peter picks an apple and realises that it is bruised. He puts the apple back in the bag and picks another one.

   a. The probability that Peter picks 2 bruised apples is:

   A. $\frac{1}{4}$ B. $\frac{1}{2}$ C. $\frac{1}{16}$ D. $\frac{3}{4}$

   b. The probability that Peter picks a bruised apple first but a good one on his second attempt is:

   A. $\frac{1}{4}$ B. $\frac{1}{2}$ C. $\frac{3}{4}$ D. $\frac{3}{16}$

9. The probability that John will be late for a meeting is $\frac{1}{7}$ and the probability that Phil will be late for a meeting is $\frac{3}{11}$.

   Calculate the probability that:

   a. John and Phil are both late
   b. neither of them is late
   c. John is late but Phil is not late
   d. Phil is late but John is not late.

10. On the roulette wheel at the casino there are 37 numbers, 0 to 36 inclusive. Bidesi puts his chip on number 8 in game 20 and on number 13 in game 21.

    a. Calculate the probability that he will win in game 20.
    b. Calculate the probability that he will win in both games.
    c. Calculate the probability that he wins at least one of the games.



11. Based on her progress through the year, Karen was given a probability of 0.8 of passing the Physics exam. If the probability of passing both Maths and Physics is 0.72, determine her probability of passing the Maths exam.

12. Suresh found that, on average, he is delayed 2 times out of 7 at Melbourne airport. Rakesh made similar observations at Brisbane airport but found that he was delayed 1 out of every 4 times.
    Determine the probability that both Suresh and Rakesh will be delayed if they are flying out of their respective airports.

13. Bronwyn has 3 pairs of Reebok and 2 pairs of Adidas running shoes. She has 2 pairs of Reebok, 3 pairs of Rio and 1 pair of Red Robin socks.
    Preparing for an early morning run, she grabs at random for a pair of socks and a pair of shoes.
    Calculate the probability that she chooses:

    a. Reebok shoes and Reebok socks
    b. Rio socks and Adidas shoes
    c. Reebok shoes and Red Robin socks
    d. Adidas shoes and socks that are not Red Robin.



14. **WE13** Two cards are drawn successively and without replacement from a pack of playing cards. Calculate the probability of drawing:

a. 2 hearts **b.** 2 kings **c.** 2 red cards.

15. In a class of 30 students there are 17 students who study Music. Two students are picked randomly to represent the class in the Student Representative Council. Calculate the probability that:

a. both students don't study Music
b. both students do study Music
c. one of the students doesn't study Music.

Reasoning

16. Greg has tossed a Tail on each of 9 successive coin tosses. He believes that his chances of tossing a Head on his next toss must be very high. Explain whether Greg is correct.

17. The Multiplication Law of probability relates to *independent events*. Tree diagrams can illustrate the sample space of successive *dependent events*, and the probability of any one combination of events can be calculated by *multiplying* the stated probabilities along the branches.
Explain whether this a contradiction to the Multiplication Law of probability.

18. Explain whether it is possible for two events, A and B, to be mutually exclusive and independent.

19. Consider the following sets:

$$\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$$

$$A = \{\text{evens in } \xi\}$$

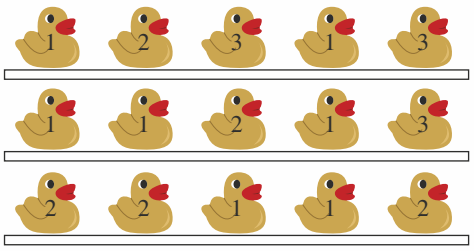
$$B = \{\text{multiples of 3 in } \xi\}$$

a. Explain whether the events A and B are independent.
b. If ξ is changed to the integers from 1 to 10 only, explain whether the result from part **a** changes.

Problem solving

20. There are three coins in a box. One coin is a fair coin, one coin is biased with an 80% chance of landing Heads, and the third is a biased coin with a 40% chance of landing Heads. A coin is selected at random and flipped.
If the result is a Head, determine the probability that the fair coin was selected.

21. A game at a carnival requires blindfolded contestants to throw balls at numbered ducks sitting on 3 shelves. The game ends when 3 ducks have been knocked off the shelves.
Assume that the probability of hitting each duck is equal.



a. Explain whether the events described in the game are dependent or independent.
b. Determine the initial probabilities of hitting each number.
c. Draw a labelled tree diagram to show the possible outcomes for a contestant.
d. Calculate the probabilities of hitting the following.

i. Pr(1, 1, 1) **ii.** Pr(2, 2, 2)
iii. Pr(3, 3, 3) **iv.** Pr(at least one 3)

22. Question **21** described a game at a carnival. A contestant pays $3 to play and must make 3 direct hits to be eligible for a prize. The numbers on the ducks hit are then summed and the contestant wins a prize according to the winners' table.

Winners' table	
Total score	**Prize**
9	Major prize ($30 value)
7 or 8	Minor prize ($10 value)
5 or 6	$2 prize
3 or 4	No prize

a. Determine the probability of winning each prize listed.
b. Suppose 1000 games are played on an average show day. Evaluate the profit that could be expected to be made by the sideshow owner on any average show day.

LESSON
11.5 Conditional probability

LEARNING INTENTIONS

At the end of this lesson you should be able to:
- determine from the language in a question whether it involves conditional probability
- use the rule for conditional probability to calculate other probabilities.

11.5.1 Recognising conditional probability

eles-4928

- **Conditional probability** is when the probability of an event is conditional on (depends on) another event occurring first.
- The effect of conditional probability is to reduce the sample space and thus increase the probability of the desired outcome.
- For two events, A and B, the conditional probability of event B, given that event A occurs, is denoted by $\Pr(B|A)$ and can be calculated using the following formula.

Probability of B given A

$$\Pr(B|A) = \frac{\Pr(A \cap B)}{\Pr(A)}, \ \Pr(A) \neq 0$$

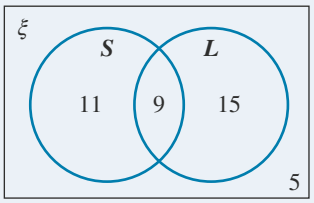
- Conditional probability can be expressed using a variety of language. Some examples of conditional probability statements follow. The key words to look for in a conditional probability statement have been highlighted in each instance.
  - **If** a student receives a B+ or better in their first Maths test, **then** the chance of them receiving a B+ or better in their second Maths test is 75%.
  - **Given that** a red marble was picked out of the bag with the first pick, the probability of a blue marble being picked out with the second pick is 0.35.
  - **Knowing that** the favourite food of a student is hot chips, the probability of their favourite condiment being tomato sauce is 68%.

WORKED EXAMPLE 14 Using a Venn diagram to calculate conditional probabilities

A group of students was asked whether they like spaghetti (S) or lasagne (L). The results are illustrated in the Venn diagram shown. Use the Venn diagram to calculate the following probabilities relating to a student's preferred food.

a. Calculate the probability that a randomly selected student likes spaghetti.

b. Determine the probability that a randomly selected student likes lasagne given that they also like spaghetti.



THINK

a. 1. Determine how many students were surveyed to identify the total number of possible outcomes. Add each of the numbers shown on the Venn diagram.

2. There are 20 students who like 'spaghetti' or 'spaghetti and lasagne', as shown in pink.

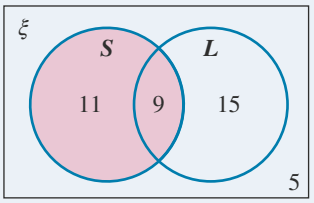
3. The probability that a randomly selected student likes spaghetti is found by substituting these values into the probability formula.

b. 1. The condition imposed 'given they also like spaghetti' alters the sample space to the 20 students described in part **a**, as shaded in blue. Of these 20 students, 9 stated that they liked lasagne and spaghetti, as shown in pink.

2. The probability that a randomly selected student likes lasagne, given that they like spaghetti, is found by substituting these values into the probability formula for conditional probability.

WRITE/DRAW

a. Total number of students $= 11 + 9 + 15 + 5 = 40$

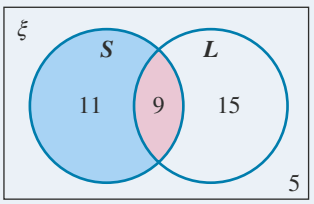


$$\Pr(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}}$$

$$\Pr(\text{spaghetti}) = \frac{20}{40}$$

$$= \frac{1}{2}$$

b.



$$\Pr(B|A) = \frac{\Pr(A \cap B)}{\Pr(A)}$$

$$\Pr(L|S) = \frac{\frac{9}{40}}{\frac{1}{2}}$$

$$= \frac{9}{20}$$

WORKED EXAMPLE 15 Using the rule for conditional probability

If $\Pr(A) = 0.3$, $\Pr(B) = 0.5$ and $\Pr(A \cup B) = 0.6$, calculate:

a. $\Pr(A \cap B)$ **b. $\Pr(B|A)$**

THINK

a. 1. State the Addition Law for probability.

WRITE

a. $\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$

2. Substitute the values given in the question into this formula and simplify.

$$0.6 = 0.3 + 0.5 - \Pr(A \cap B)$$
$$\Pr(A \cap B) = 0.3 + 0.5 - 0.6$$
$$= 0.2$$

b. 1. State the formula for conditional probability.

b. $\Pr(B|A) = \dfrac{\Pr(A \cap B)}{\Pr(A)}, \Pr(A) \neq 0$

2. Substitute the values given in the question into this formula and simplify.

$$\Pr(B|A) = \frac{0.2}{0.3}$$
$$= \frac{2}{3}$$

- It is possible to transpose the formula for conditional probability to calculate $\Pr(A \cap B)$:

$$\Pr(B|A) = \frac{\Pr(A \cap B)}{\Pr(A)}, \Pr(A) \neq 0$$
$$\Pr(A \cap B) = \Pr(A) \times \Pr(B|A)$$

Resources

eWorkbook Topic 11 Workbook (worksheets, code puzzle and project) (ewbk-2103)
Video eLesson Conditional probability (eles-1928)
Interactivities Individual pathway interactivity: Conditional probability (int-4619)
Conditional probability (int-6085)

Exercise 11.5 Conditional probability

learnon

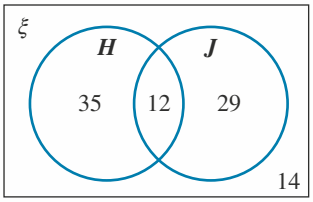
11.5 Quick quiz | 11.5 Exercise

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 3, 5, 10, 12, 15	2, 6, 9, 13, 16	4, 7, 8, 11, 14, 17

Fluency

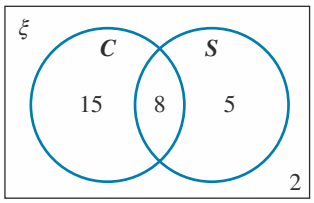
1. WE14 A group of students was asked whether they liked the following forms of dance: hip hop (H) or jazz (J). The results are illustrated in the Venn diagram.



Use the Venn diagram to calculate the following probabilities relating to a student's favourite form of dance.

a. Calculate the probability that a randomly selected student likes jazz.
b. Determine the probability that a randomly selected student likes hip hop, given that they like jazz.

2. A group of students was asked which seats they liked: the seats in the computer lab or the seats in the science lab. The results are illustrated in the Venn diagram.



Use the Venn diagram to calculate the following probabilities relating to the most comfortable seats.

a. Calculate the probability that a randomly selected student likes the seats in the science lab.

b. Determine the probability that a randomly selected student likes the seats in the science lab, given that they like the seats in the computer lab or the science lab.

3. **WE15** If $\Pr(A) = 0.7$, $\Pr(B) = 0.5$ and $\Pr(A \cup B) = 0.9$, calculate:

a. $\Pr(A \cap B)$ **b.** $\Pr(B|A)$.

4. If $\Pr(A) = 0.65$, $\Pr(B) = 0.75$ and $\Pr(A \cap B) = 0.45$, calculate:

a. $\Pr(B|A)$ **b.** $\Pr(A|B)$.

Understanding

5. A medical degree requires applicants to participate in two tests: an aptitude test and an emotional maturity test. This year 52% passed the aptitude test and 30% passed both tests. Use the conditional probability formula to calculate the probability that a student who passed the aptitude test also passed the emotional maturity test.

6. At a school classified as a 'Music school for excellence', the probability that a student elects to study Music and Physics is 0.2. The probability that a student takes Music is 0.92.
Determine the probability that a student takes Physics, given that the student is taking Music.



7. The probability that a student is well and misses a work shift the night before an exam is 0.045, and the probability that a student misses a work shift is 0.05.
Determine the probability that a student is well, given they miss a work shift the night before an exam.

8. Two marbles are chosen, without replacement, from a jar containing only red and green marbles. The probability of selecting a green marble and then a red marble is 0.1641. The probability of selecting a green marble on the first draw is 0.8.
Determine the probability of selecting a red marble on the second draw, given the first marble drawn was green.



9. MC A group of 80 schoolgirls consists of 54 dancers and 35 singers. Each member of the group is either a dancer, a singer, or both. The probability that a randomly selected student is a singer given that she is a dancer is:

A. 0.17 **B.** 0.44 **C.** 0.68 **D.** 0.11

10. Consider rolling a red and a black die and the probabilities of the following events:

Event A the red die lands on 5
Event B the black die lands on 2
Event C the sum of the dice is 10.

a. MC The initial probability of each event described is:

A. $\Pr(A) = \frac{1}{6}$, $\Pr(B) = \frac{1}{6}$, $\Pr(C) = \frac{1}{6}$

B. $\Pr(A) = \frac{5}{6}$, $\Pr(B) = \frac{2}{6}$, $\Pr(C) = \frac{7}{36}$

C. $\Pr(A) = \frac{5}{6}$, $\Pr(B) = \frac{2}{6}$, $\Pr(C) = \frac{5}{18}$

D. $\Pr(A) = \frac{1}{6}$, $\Pr(B) = \frac{1}{6}$, $\Pr(C) = \frac{1}{12}$

b. Calculate the following probabilities.

i. $\Pr(A|B)$ **ii.** $\Pr(B|A)$ **iii.** $\Pr(C|A)$ **iv.** $\Pr(C|B)$

11. The following is the blood pressure data from 232 adult patients admitted to a hospital over a week. The results are displayed in a two-way frequency table.

Age	Blood pressure			Total
	Low	Medium	High	
Under 60 years	92	44	10	146
60 years or above	17	46	23	86
Total	109	90	33	**232**

a. Calculate the probability that a randomly chosen patient has low blood pressure.
b. Determine the probability that a randomly chosen patient is under 60 years of age.
c. Calculate the probability that a randomly chosen patient has high blood pressure, given they are aged 60 years or above.
d. Determine the probability that a randomly chosen patient is under the age of 60, given they have medium blood pressure.

Reasoning

12. Explain how imposing a condition alters probability calculations.

13. At a school, 65% of the students are male and 35% are female. Of the male students, 10% report that dancing is their favourite activity; of the female students, 25% report that dancing is their favourite activity.



Determine the probability that:

a. a student selected at random prefers dancing and is female
b. a student selected at random prefers dancing and is male.
c. Construct a tree diagram to present the information given and use it to calculate:
  i. the probability that a student is male and does not prefer dancing
  ii. the overall percentage of students who prefer dancing.

14. Consider the following sets of numbers.

$$\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$$

$$A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$B = \{2, 3, 5, 7, 11, 13, 17, 19\}$$

a. Calculate the following probabilities.

i. $\Pr(A)$

ii. $\Pr(B)$

iii. $\Pr(A \cap B)$

iv. $\Pr(A|B)$

v. $\Pr(B|A)$

b. Explain whether the events A and B are independent.

c. Write a statement that connects $\Pr(A)$, $\Pr(A \cap B)$, $\Pr(A|B)$ and independent events.

Problem solving

15. The rapid test used to determine whether a person is infected with COVID-19 is not perfect. For one type of rapid test, the probability of a person with the disease returning a positive result is 0.98, while the probability of a person without the disease returning a positive result is 0.04.
At its peak in a certain country, the probability that a randomly selected person has COVID-19 is 0.05.
Determine the probability that a randomly selected person will return a positive result.

16. Two marbles are chosen, without replacement, from a jar containing only red and green marbles. The probability of selecting a green marble and then a red marble is 0.72. The probability of selecting a green marble on the first draw is 0.85.
Determine the probability of selecting a red marble on the second draw if the first marble drawn was green.

17. When walking home from school during the summer months, Harold buys either an ice-cream or a drink from the corner shop.
If Harold bought an ice-cream the previous day, there is a 30% chance that he will buy a drink the next day. If he bought a drink the previous day, there is a 40% chance that he will buy an ice-cream the next day. On Monday, Harold bought an ice-cream. Determine the probability that he buys an ice-cream on Wednesday.



LESSON
11.6 Counting principles and factorial notation (10A)

LEARNING INTENTIONS

At the end of this lesson you should be able to:

- apply the multiplication principle as a counting technique
- calculate permutations using factorials
- calculate permutations for situations involving like objects
- calculate the number of possible combinations in situations where order is not important.

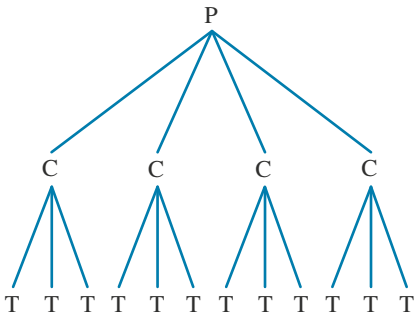
11.6.1 Counting techniques

eles-6143

- Counting techniques allow us to determine the number of ways an activity can occur. This in turn allows us to calculate the probability of an event.
- When considering a set with n elements, the number of different subsets formed can be found by considering and systematically listing each element for inclusion.
  For example, the set $\{a, b, c\}$ has subsets $\{\varnothing, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$.
- In the above example there are 8 subsets. This could have also been calculated by $2^n = 2^3$, where n is the number of elements in the set.
- Tree diagrams can be used to determine the number of solutions to simple counting problems.
  For example, a tree diagram can be made to answer the following question:

  If driving from Port Douglas to Cairns I can take any one of 4 different roads and from Cairns to Townsville I can take 3 different roads, determine how many different routes I can take if I want to drive from Port Douglas to Townsville.

- There are four ways of getting from Port Douglas to Cairns and three ways of getting from Cairns to Townsville, so there are $4 \times 3 = 12$ ways of getting from Port Douglas to Townsville.
- This is known as the **multiplication principle** and should be used when there are events where one event is followed by the other — that is, when order is important.



The multiplication principle

If there are n ways of performing operation A and m ways of performing operation B, then there are $n \times m$ ways of performing A *and* B in the order AB.

- A useful technique for solving problems based on the multiplication principle is to use boxes. In the example above we would write the following.

1st	2nd
4	3

- The value in the '1st' box represents the number of ways the first operation can be performed.
- The value in the '2nd' box stands for the number of ways the second operation can be performed.
- To apply the multiplication principle you multiply the numbers in the boxes.

WORKED EXAMPLE 16 Applying the multiplication principle

Two letters are to be chosen from the set of 5 letters A, B, C, D and E, where order is important.

a. List all the different ways that this may be done.

b. Use the multiplication principle to calculate the number of ways that this may be done.

c. Determine the probability the first letter will be a C.

d. Use the multiplication principle to calculate the number of ways the two letters can be chosen if the first letter is replaced after selection.

THINK	WRITE
a. 1. Begin with A in first place and make a list of each of the possible pairs.	**a.** AB AC AD AE
2. Make a list of each of the possible pairs with B in the first position.	BA BC BD BE
3. Make a list of each of the possible pairs with C in the first position.	CA CB CD CE
4. Make a list of each of the possible pairs with D in the first position.	DA DB DC DE
5. Make a list of each of the possible pairs with E in the first position. *Note:* AB and BA need to be listed separately as order is important.	EA EB EC ED
b. The multiplication principle could have been used to determine the number of ordered pairs.	**b.**
1. Rule up two boxes that represent the pair.	
2. Write down the number of letters that may be selected for the first box. That is, in first place any of the 5 letters may be used.	
3. Write down the number of letters that may be selected for the second box. That is, in second place any of the 4 letters may be used. *Note:* One less letter is used to avoid repetition.	
4. Evaluate.	$5 \times 4 = 20$ ways
5. Write the answer.	There are 20 ways in which 2 letters may be selected from a group of 5 where order is important.
c. 1. Recall the probability formula. The total number in the set $n(\xi)$ was determined in part **b**.	**c.** $\Pr(A) = \frac{n(A)}{n(\xi)}$ $n(\xi) = 20$
2. Let A be the event that the pair starts with a C. Draw a table showing the requirement imposed by the first letter to be C.	

5	4

1	

3. Complete the table. Once the first letter has been completed, there are 4 choices for the second letter. Use the multiplication principle to determine the number of combinations starting with C.

1	4

There are $1 \times 4 = 4$ possible combinations beginning with C.
So, $n(A) = 4$.

4. Use the probability formula to determine the probability the first letter will be a C.

$$\Pr(A) = \frac{n(A)}{n(\xi)} = \frac{4}{20} = \frac{1}{5}$$

This is confirmed by examining the answer to part **a**.

d. 1. Rule up two boxes that represent the pair.

d.

5	5

2. Write down the number of letters that may be selected for the first box. That is, in first place any of the 5 letters may be used.
3. Write down the number of letters that may be selected for the second box. That is, in second place, any of the 5 letters may be used.
Note: The first letter has been replaced so there are 5 to choose from again.
4. Evaluate.
5. Write the answer.

$5 \times 5 = 25$ ways

There are 25 ways in which 2 letters may be selected from a group of 5 with replacement.

WORKED EXAMPLE 17 Applying the multiplication principle to determine the number of ways and probability of an activity

a. Use the multiplication principle to calculate how many ways an arrangement of 5 numbers can be chosen from {1, 2, 3, 4, 5, 6}.
b. Determine the probability of the number ending with 4.

THINK

a. 1. Instead of listing all possibilities, draw 5 boxes to represent the 5 numbers chosen.
Label each box on the top row as 1st, 2nd, 3rd, 4th and 5th.
Note: The word 'arrangement' implies order is important.

WRITE

a.

2. Fill in each of the boxes showing the number of ways a number may be chosen.
In the 1st box there are 6 choices for the first number. In the 2nd box there are 5 choices for the second number as 1 number has already been used. In the 3rd box there are 4 choices for the third number as 2 numbers have already been used. Continue this process until each of the 5 boxes is filled.

1st	2nd	3rd	4th	5th
6	5	4	3	2

3. Use the multiplication principle as this is an '*and*' situation.

Number of ways $= 6 \times 5 \times 4 \times 3 \times 2$
$= 720$

4. Write the answer.

An arrangement of 5 numbers may be chosen 720 ways.

b. 1. Recall the probability formula.
The total number of arrangements, $n(\xi)$, was determined in part **a**.

b. $\Pr(A) = \dfrac{n(A)}{n(\xi)}$
$n(\xi) = 720$

2. Let A be the event that the number ends with 4.
Draw a table showing the requirement imposed by the last letter to be 4.

1st	2nd	3rd	4th	5th
				1

3. Complete the table. Once the last number has been completed, there are 5 choices for the number in the first position and 4 choices for the next number. Continue this process until each of the 5 columns has been filled. Use the multiplication principle to determine the number of combinations ending with 4.

1st	2nd	3rd	4th	5th
5	4	3	2	1

There are $5 \times 4 \times 3 \times 2 \times 1 = 120$ possible combinations ending with 4.
So, $n(A) = 120$.

4. Use the probability formula to determine the probability of the number ending with 4 .

$$\Pr(A) = \frac{n(A)}{n(\xi)}$$
$$= \frac{120}{720}$$
$$= \frac{1}{6}$$

11.6.2 Factorials

eles-6144

- The multiplication principle can be used to determine the number of ways in which five basketball championship trophies could be displayed on a shelf in the school foyer.
- The display may be done in the following way:

Position 1	Position 2	Position 3	Position 4	Position 5
5	4	3	2	1

That is, there are $5 \times 4 \times 3 \times 2 \times 1 = 120$ ways.

- If there were a large number of items to arrange, this method could be time consuming, but there is a more efficient method using **factorials**.
- In general when we need to multiply each of the integers from a particular number, n, down to 1, we write $n!$, which is read as 'n factorial'.
  Hence:

$$\begin{aligned} 6! &= 6 \times 5 \times 4 \times 3 \times 2 \times 1 \\ &= 720 \\ 8! &= 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 \\ &= 40\,320 \\ n! &= n \times (n-1) \times (n-2) \times (n-3) \times \ldots \times 3 \times 2 \times 1 \end{aligned}$$

- The number of ways n distinct objects may be arranged is $n!$ (n factorial).

Factorials

$$n! = n \times (n-1) \times (n-2) \times (n-3) \times \ldots \times 3 \times 2 \times 1$$

That is, $n!$ is the product of each of the integers from n down to 1.

A special case of the factorial function is $0! = 1$.

WORKED EXAMPLE 18 Evaluating factorials

Evaluate the following factorials.

a. $4!$ **b.** $7!$ **c.** $\frac{8!}{5!}$

THINK

a. 1. Write 4! in its expanded form and evaluate.

2. Verify the answer obtained using the factorial function on a calculator.

b. 1. Write 7! in its expanded form and evaluate.

2. Verify the answer obtained using the factorial function on a calculator.

c. 1. Write each factorial term in its expanded form.

2. Cancel down like terms.

3. Evaluate.

4. Verify the answer obtained using the factorial function on a calculator.

WRITE

a. $4! = 4 \times 3 \times 2 \times 1$
$= 24$

b. $7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$
$= 5040$

c. $\frac{8!}{5!} = \frac{8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{5 \times 4 \times 3 \times 2 \times 1}$
$= 8 \times 7 \times 6$
$= 336$

TI | THINK

c. On a Calculator page, complete the entry line as shown.

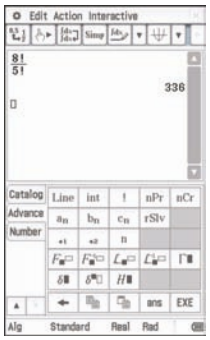
DISPLAY/WRITE



CASIO | THINK

c. From the Advance palette on the Keyboard, complete the entry line as shown by selecting !.

DISPLAY/WRITE



11.6.3 Permutations

eles-6145

- When objects are arranged in a particular order, it is called an arrangement or **permutation**.
- Previously, we learned that if you select 3 letters from 7 where the *order is important*, the number of possible arrangements can be found using the multiplication principle.

1st	2nd	3rd
7	6	5

$$\begin{aligned}\text{The number of arrangements} &= 7\times6\times5\\ &= 210\end{aligned}$$

- This could also be expressed in factorial form:

$$\begin{aligned}7\times6\times5 &= \frac{7\times6\times5\times4\times3\times2\times1}{4\times3\times2\times1}\\ &= \frac{7!}{4!}\\ &= \frac{7!}{(7-3)!}\end{aligned}$$

- Using more formal terminology we say that in choosing 3 things from 7 things where order is important, the number of permutations is ${}^{7}\mathrm{P}_{3} = 7\times6\times5$. The letter P is used to remind us that we are finding permutations.
- The number of ways of choosing $\boldsymbol{r}$ things from $\boldsymbol{n}$ distinct things is given by the following rule.

Permutations

$$^{n}\mathrm{P}_{r} = \frac{n!}{(n-r)!}$$

WORKED EXAMPLE 19 Determining the number of ways an activity can be arranged

Courtney has purchased 9 new books but only has room for 3 of them on her bookshelf. Determine how many ways the books can be arranged.

THINK

1. Recall the rule for permutations.

WRITE

$$^{n}\mathrm{P}_{r} = \frac{n!}{(n-r)!}$$

2. Substitute the given values of n and r into the permutation formula.

$$^9P_3 = \frac{9!}{(9-3)!}$$
$$= \frac{9!}{6!}$$

3. Use a calculator to evaluate 9! and 6!.

$$= 504$$

4. Write the answer.

There are 504 ways the books can be arranged.

WORKED EXAMPLE 20 Using permutations to calculate probabilities

The netball club needs to appoint a president, secretary and treasurer. From the committee 7 people have volunteered for these positions. Each of the 7 nominees is happy to fill any one of the 3 positions.

a. Determine how many different ways these positions can be filled.

b. For three years, the same 7 people volunteer for these positions. Determine the probability one of them is president 3 years in a row.

THINK	WRITE
a. 1. Recall the rule for permutations. *Note:* Order is important, so use permutations.	**a.** $^nP_r = \frac{n!}{(n-r)!}$
2. Substitute the given values of n and r into the permutation formula.	$^7P_3 = \frac{7!}{(7-3)!}$ $= \frac{7!}{4!}$
3. Use a calculator to evaluate 7! and 4!.	$= \frac{5040}{24}$
4. Evaluate.	$= 210$
5. Write the answer.	There are 210 different ways of filling the positions of president, secretary and treasurer.
b. 1. In three years of the president's position, each year this could be awarded to one of the 7 people.	**b.** 7 | 7 | 7 The total number of ways the president's position could be filled is $7 \times 7 \times 7$. $n(\xi) = 7 \times 7 \times 7$ $= 343$
2. Let A be the event that the same person fills the position three years in a row.	There are 7 choices of the same person to fill the positions three years in a row. $n(A) = 7$
3. Calculate the probability that the same person fills the president's position three years in a row.	$P(A) = \frac{n(A)}{n(\xi)}$ $= \frac{7}{7 \times 7 \times 7}$ $= \frac{1}{49}$ The probability that one of the 7 volunteers fills the president's position three years in a row is $\frac{1}{49}$.

11.6.4 Permutations with like objects

eles-6146

- If some objects are repeated or are like objects, it will decrease the number of different arrangements. For example, a 5-letter word made from 5 different letters would have $5! = 120$ arrangements, but if the letters were 3 A's and 2 B's the number of arrangements would be less.
- To analyse the situation let us imagine that we can distinguish one A from another. We will write A_1, A_2, A_3, B_1 and B_2 to represent the 5 letters.
- As we list some of the possible arrangements we notice that some are actually the same, as shown in the table.

$A_1A_2B_1A_3B_2$ $A_1A_3B_1A_2B_2$ $A_2A_1B_1A_3B_2$ $A_2A_3B_1A_1B_2$ $A_3A_1B_1A_2B_2$ $A_3A_2B_1A_1B_2$	$A_1A_2B_2A_3B_1$ $A_1A_3B_2A_2B_1$ $A_2A_1B_2A_3B_1$ $A_2A_3B_2A_1B_1$ $A_3A_1B_2A_2B_1$ $A_3A_2B_2A_1B_1$	Each of these 12 arrangements is the same — AABAB — if $A_1 = A_2 = A_3$ and $B_1 = B_2$.
$B_2A_1A_2B_1A_3$ $B_2A_1A_3B_1A_2$ $B_2A_2A_1B_1A_3$ $B_2A_2A_3B_1A_1$ $B_2A_3A_1B_1A_2$ $B_2A_3A_2B_1A_1$	$B_1A_1A_2B_2A_3$ $B_1A_1A_3B_2A_2$ $B_1A_2A_1B_2A_3$ $B_1A_2A_3B_2A_1$ $B_1A_3A_1B_2A_2$ $B_1A_3A_2B_2A_1$	Each of these 12 arrangements is the same — BAABA — if $A_1 = A_2 = A_3$ and $B_1 = B_2$.

- The number of repetitions is 3! for the As and 2! for the Bs. Thus, the number of different arrangements in choosing 5 letters from 3 As and 2 Bs is $\frac{5!}{3! \times 2!} = 10$.

Permutations with like objects

The number of different ways of arranging n objects made up of groups of repeated (identical) objects, n_1 in the first group, n_2 in the second group and so on, is:

$$\frac{n!}{n_1!\, n_2!\, n_3! \ldots n_r!}.$$

WORKED EXAMPLE 21 Calculating permutations with like objects

Determine how many different permutations of 7 counters can be made from 4 black and 3 white counters.

THINK

1. Write down the total number of counters.
2. Write down the number of times any of the coloured counters are repeated.
3. Write down the formula for arranging groups of like things.
4. Substitute the values of n, n_1 and n_2 into the rule.

WRITE

There are 7 counters in all; therefore, $n = 7$.

There are 3 white counters; therefore, $n_1 = 3$. There are 4 black counters; therefore, $n_2 = 4$.

$$\frac{n!}{n_1!\, n_2!\, n_3! \ldots n_r!}$$

$$= \frac{7!}{3! \times 4!}$$

5. Expand each of the factorials. $= \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1 \times 4 \times 3 \times 2 \times 1}$

6. Simplify the fraction. $= \frac{7 \times 6 \times 5}{6}$

7. Evaluate. $= 35$

8. Write the answer. Thirty-five different arrangements can be made from 7 counters, of which 3 are white and 4 are black.

11.6.5 Combinations

eles-6147

- A group of things chosen from a larger group where order is not important is called a **combination**.
- Suppose 5 people have nominated for a committee consisting of 3 members. It does not matter in what order the candidates are placed on the committee, it matters only whether they are there or not.
- If order was important we know there would be ${}^5P_3$, or 60, ways in which this could be done. Here are the possibilities:

ABC	ACB	BAC	BCA	CAB	CBA
ABD	ADB	BAD	BDA	DAB	DBA
ABE	AEB	BAE	BEA	EAB	ABA
ACE	AEC	CAE	CEA	EAC	ECA
ACD	ADC	CAD	CDA	DAC	DCA
ADE	AED	DAE	DEA	EAD	EDA
BCD	BDC	CBD	CDB	DBC	DCB
BCE	BEC	CBE	CEB	EBC	ECB
BDE	BED	DBE	DEB	EBD	EDB
CDE	CED	DCE	DEC	ECD	EDC



- The 60 arrangements are different only if we take order into account; that is, ABC is different from CAB and so on. You will notice in this table that there are 10 distinct committees corresponding to the 10 distinct rows. Each row merely repeats, in a different order, the committee in the first column. This result (10 distinct committees) can be arrived at logically:
  1. There are ${}^5P_3$ ways of choosing or selecting 3 from 5 in order.
  2. Each choice of 3 is repeated 3! times.
  3. The number of distinct selections or combinations is ${}^5P_3 \div 3! = 10$.
- This leads to the general rule of selecting r objects from n objects.

Combinations

The number of ways of choosing or selecting r objects from n distinct objects, where order is not important, is given by ${}^nC_r$:

$$ {}^nC_r = \frac{{}^nP_r}{r!} $$

C is used to represent combinations.

WORKED EXAMPLE 22 Evaluating combinations

Write these combinations as statements involving permutations and then calculate them.

a. ${}^{7}C_{2}$ **b.** ${}^{20}C_{3}$

THINK	WRITE
a. 1. Recall the rule for ${}^{n}C_{r}$.	**a.** ${}^{n}C_{r} = \frac{{}^{n}P_{r}}{r!}$
2. Substitute the given values of n and r into the combination formula.	${}^{7}C_{2} = \frac{{}^{7}P_{2}}{2!}$
3. Simplify the fraction.	$= \frac{\left(\frac{7!}{5!}\right)}{2!}$ $= \frac{7!}{5!} \div 2!$ $= \frac{7!}{5!} \times \frac{1}{2!}$ $= \frac{7 \times 6 \times 5!}{5! \times 2 \times 1}$
4. Evaluate.	$= \frac{7 \times 6}{2 \times 1}$ $= \frac{42}{2}$ $= 21$
b. 1. Write down the rule for ${}^{n}C_{r}$.	**b.** ${}^{n}C_{r} = \frac{{}^{n}P_{r}}{r!}$
2. Substitute the values of n and r into the formula.	${}^{20}C_{3} = \frac{{}^{20}P_{3}}{3!}$
3. Simplify the fraction.	$= \frac{\left(\frac{20!}{17!}\right)}{3!}$ $= \frac{20!}{17!} \div 3!$ $= \frac{20!}{17!} \times \frac{1}{3!}$ $= \frac{20 \times 19 \times 18 \times 17!}{17! \times 3 \times 2 \times 1}$
4. Evaluate.	$= \frac{20 \times 19 \times 18}{3 \times 2 \times 1}$ $= \frac{6840}{6}$ $= 1140$

WORKED EXAMPLE 23 Applying combinations

Apply the concept of ${}^nC_r$ to calculate the number ways a basketball team of 5 players be selected from a squad of 9 if the order in which they are selected does not matter.



THINK

1. Recall the rule for ${}^nC_r$.
   Note: Since order does not matter, use the ${}^nC_r$ rule.
2. Substitute the values of n and r into the formula.
3. Simplify the fraction.
4. Evaluate.

WRITE

$${}^nC_r = \frac{{}^nP_r}{r!}$$

$${}^9C_5 = \frac{{}^9P_5}{5!}$$

$$= \frac{\left(\frac{9!}{4!}\right)}{5!}$$

$$= \frac{9!}{4!} \div 5!$$

$$= \frac{9!}{4!} \times \frac{1}{5!}$$

$$= \frac{9!}{4!\ 5!}$$

$$= \frac{9 \times 8 \times 7 \times 6 \times 5!}{4 \times 3 \times 2 \times 1 \times 5!}$$

$$= \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1}$$

$$= \frac{3024}{24}$$

$$= 126$$

- To calculate probabilities using combinations, express the numerator and denominator in terms of the appropriate coefficients and then carry out the calculations.

WORKED EXAMPLE 24 Calculating probabilities using combinations

A committee of 5 students is to be chosen from 7 boys and 4 girls. Use ${}^nC_r$ and the multiplication and addition principles to answer the following.

a. Calculate how many committees can be formed.

b. Calculate how many of the committees contain exactly 2 boys and 3 girls.

THINK	WRITE
a. 1. As there is no restriction, choose the committee from the total number of students.	**a.** There are 11 students in total from whom 5 students are to be chosen. This can be done in ${}^{11}C_5$ ways.
2. Use the formula ${}^{n}C_{r} = \frac{n!}{r! \times (n-r)!}$ to calculate the answer.	${}^{11}C_5 = \frac{11!}{5! \times (11 - 5)!}$ $= \frac{11!}{5! \times 6!}$ $= \frac{11 \times 10 \times 9 \times 8 \times 7 \times \cancel{6!}}{5! \times \cancel{6!}}$ $= \frac{11 \times 10 \times 9 \times 8 \times 7}{5 \times 4 \times 3 \times 2 \times 1}$ $= 462$ There are 462 possible committees.
b. 1. Select the committee to satisfy the given restriction.	**b.** The 2 boys can be chosen from the 7 boys available in ${}^{7}C_2$ ways. The 3 girls can be chosen from the 4 girls available in ${}^{4}C_3$ ways. The total number of committees that contain two boys and three girls is ${}^{7}C_2 \times {}^{4}C_3$.
2. Use the multiplication principle to form the total number of committees. *Note:* The upper numbers on the combinatoric coefficients sum to the total available, $7 + 4 = 11$, and the lower numbers sum to the number that must be on the committee, $2 + 3 = 5$.	
3. Calculate the answer.	${}^{7}C_2 \times {}^{4}C_3 = \frac{7!}{2! \times 5!} \times 4$ $= \frac{7 \times 6}{2!} \times 4$ $= 21 \times 4$ $= 84$ There are 84 committees possible with the given restriction.

Resources

eWorkbook Topic 11 Workbook (worksheets, code puzzle and project) (ewbk-2103)

Exercise 11.6 Counting principles and factorial notation (10A)

learnon

11.6 Quick quiz on | 11.6 Exercise

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 4, 7, 12, 14, 15, 17, 22, 27	2, 5, 8, 10, 13, 16, 18, 20, 23, 24, 28, 29	3, 6, 9, 11, 19, 21, 25, 26, 30, 31, 32

Fluency

1. WE16 Two letters are to be chosen from A, B and C, where order is important.
   a. List all the different ways that this may be done.
   b. Use the multiplication principle to calculate the number of ways that this may be done.
   c. Determine the probability the last letter will be a B.
   d. Use the multiplication principle to calculate the number of ways the two letters can be chosen if the first letter is replaced after selection.
2. List all the different arrangements possible for a group of 2 colours to be chosen from B (blue), G (green), Y (yellow) and R (red).
3. List all the different arrangements possible for a group of 3 letters to be chosen from A, B and C.
4. a. WE17 Use the multiplication principle to calculate how many ways an arrangement of 2 letters can be chosen from A, B, C, D, E, F and G.
   b. Calculate how many ways an arrangement of 3 letters can be chosen from 7 different letters.
   c. Calclate how many ways an arrangement of 4 letters can be chosen from 7 different letters.
   d. Calculate how many different arrangements of 5 letters can be made from 7 letters.
   e. Determine how many possible combinations there are beginning with an E.
5. Determine the number of different subsets formed from a set with the following numbers of elements.
   a. 3 b. 5 c. 9
6. Recall the definition of $n!$ and write each of the following in expanded form.
   a. 4! b. 5! c. 6! d. 7!
7. WE18 Evaluate the following factorials without using a calculator.
   a. $\frac{9!}{5!}$ b. $\frac{10!}{4!}$ c. $\frac{7!}{3!}$ d. $\frac{6!}{0!}$
8. Evaluate the following factorials.
   a. 10! b. 14! c. 9! d. 16!
9. Calculate each of the following by expressing it in expanded form.
   a. ${}^{8}P_{2}$ b. ${}^{7}P_{5}$ c. ${}^{8}P_{7}$
10. Write each of the following as a quotient of factorials and hence evaluate.
    a. ${}^{9}P_{6}$ b. ${}^{5}P_{2}$ c. ${}^{18}P_{5}$
11. Use your calculator to determine the value of each of the following.
    a. ${}^{20}P_{6}$ b. ${}^{800}P_{2}$ c. ${}^{18}P_{5}$

12. **WE 21** Determine how many different arrangements of 5 counters can be made using 3 red and 2 blue counters.

13. Determine how many different arrangements of 9 counters can be made using 4 black, 3 red and 2 blue counters.

14. **WE 22** Write each of the following as statements in terms of permutations.

a. ${}^{8}C_{3}$ **b.** ${}^{19}C_{2}$ **c.** ${}^{1}C_{1}$ **d.** ${}^{5}C_{0}$

Understanding

15. **a.** A teddy bear's wardrobe consists of 3 different hats, 4 different shirts and 2 different pairs of pants. Determine how many different outfits the teddy bear can wear.

b. A surfboard is to have 1 colour on its top and a different colour on its bottom. The 3 possible colours are red, blue and green.
Calculate how many different ways the surfboard can be coloured.

c. A new phone comes with a choice of 3 cases, 2 different-sized screens and 2 different storage capacities. With these choices, determine how many different arrangements are possible.

d. Messages can be sent by placing 3 different coloured flags in order on a pole. If the flags come in 4 colours, determine how many different messages can be sent.

16. Hani and Mary's restaurant offers its patrons a choice of 4 entrees,10 main courses and 5 desserts.

a. Determine how many choices of 3-course meals (entree, main, dessert) are available.

b. Determine how many choices of entree and main course are offered.

c. Determine how many choices of meals comprising a main course and dessert are offered.

17. **WE20** A soccer club will appoint a president and a vice-president. Eight people have volunteered for either of the two positions.

a. Determine how many different ways these positions can be filled.

b. For two years the same 8 people volunteered for these positions.
Determine the probability that one of them is president for both years.

18. Determine how many different permutations can be made using the 11 letters of the word ABRACADABRA.

19. A collection of 12 books is to be arranged on a shelf.



The books consist of 3 copies of *Great Expectations*, 5 copies of *Catcher in the Rye* and 4 copies of *Huntin', Fishin' and Shootin'*. Determine how many different arrangements of these books are possible.

20. **WE23** Apply the concept of ${}^{n}C_{r}$ to calculate the number of ways three types of ice-cream can be chosen in any order from a supermarket freezer if the freezer contains the following numbers of types of ice-cream.

a. 3 types **b.** 6 types **c.** 10 types **d.** 12 types

21. A mixed netball team must have 3 women and 4 men in the side. If the squad has 6 women and 5 men wanting to play, determine how many different teams are possible.

Reasoning



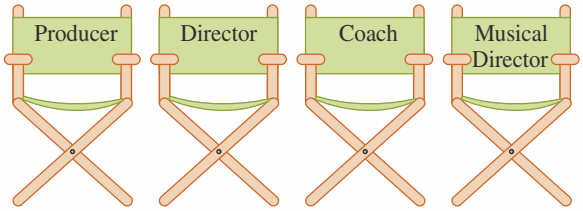
22. Jasmin has a phone that has a 4-digit security code. She remembers that the first number in the code was 9 and that the others were 3, 4 and 7 but forgets the order of the last 3 digits.
Determine how many different trials she must make to be sure of unlocking the phone.

23. Determine how many different 4-digit numbers can be made from the numbers 1, 3, 5 and 7 if the numbers can be repeated (that is 3355 and 7777 are valid).

24. There are 26 players in an online game. Determine how many different results for 1st, 2nd, 3rd and 4th can occur.

25. A rowing crew consists of 4 rowers who sit in a definite order. Determine how many different crews are possible if 5 people try out for selection.

26. The school musical needs a producer, director, musical director and script coach. Nine people have volunteered for any of these positions.



Determine how many different ways the positions can be filled. (*Note:* One person cannot take on more than 1 position.)

Problem solving

27. A shelf holding 24 cans of dog food is to be stacked using 9 cans of Yummy and 15 cans of Ruff for Dogs. Determine how many different ways the shelf can be stocked.

28. WE24 A committee of 5 students is to be chosen from 6 boys and 8 girls. Use ${}^nC_r$ and the multiplication principle to answer the following questions.

a. Calculate how many committees can be formed.
b. Calculate how many of the committees contain exactly 2 boys and 3 girls.

29. **a.** Calculate the values of:

i. ${}^{12}C_3$ and ${}^{12}C_9$
ii. ${}^{15}C_8$ and ${}^{15}C_7$
iii. ${}^{10}C_1$ and ${}^{10}C_9$
iv. ${}^{8}C_3$ and ${}^{8}C_5$

b. Describe what you notice in your results for part **a**.
Give your answer as a general statement such as 'The value of ${}^nC_r$ is...'.

30. Lotto is a game played by choosing 6 numbers from 45. Players try to match their choice with those numbers chosen at the official draw. No number can be drawn more than once, and the order in which the numbers are selected does not matter.



a. Calculate how many different selections of 6 numbers can be made from 45.
b. Suppose the first numbers drawn at the official draw are 42, 3 and 18.
Determine how many selections of 6 numbers will contain these 3 numbers.

31. A bag contains 7 red and 3 white balls. A ball is taken at random, its colour is noted, and it is then replaced before a second ball is chosen at random and its colour is noted.

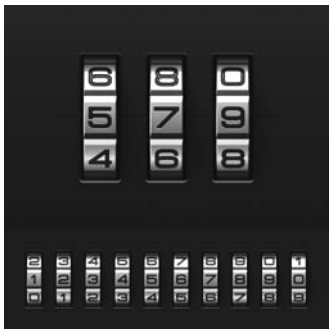


a. Show the possible outcomes with a fully labelled tree diagram.
b. As the first ball was chosen, how many balls were in the bag?
c. As the second ball was chosen, how many balls were in the bag?
d. Does the probability of choosing a red or white ball change from the first selection to the second? Explain.
e. Calculate the probability of choosing a red ball twice.

The process is repeated. However, this time the first selection is not replaced.

f. As the second ball is chosen, how many balls are in the bag?
g. Does the probability of choosing a red or white ball change from the first selection to the second? Explain.
h. Construct a fully labelled tree diagram to show all possible outcomes.
i. Calculate the probability of choosing two red balls.

32. A combination lock has 3 digits, each from 0 to 9.



a. Determine how many combinations are possible.

The lock mechanism becomes loose and will open if the digits are within 1 either side of the correct digit. For example, if the true combination is 382, then the lock will open on 271, 272, 371, 493 and so on.

b. Determine how many combinations would unlock the safe.
c. List the possible combinations that would open the lock if the true combination is 382.

LESSON
11.7 Review

11.7.1 Topic summary

PROBABILITY

Review of probability

- The sample space is the set of all possible outcomes. It is denoted ξ. For example, the sample space of rolling a die is $\xi = \{1, 2, 3, 4, 5, 6\}$.
- An event is a set of favourable outcomes. For example, if A is the event {rolling an even number on a die}, $A = \{2, 4, 6\}$.
- The probability of an outcome or event is always between 0 and 1.
- The sum of all probabilities in a sample space is 1.
- An event that is **certain** has a probability of 1.
- An event that is **impossible** has a probability of 0.
- When all outcomes are equally likely the theoretical probability of an event is given by the following formula:

$$\Pr(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}} = \frac{n(E)}{n(\xi)}$$

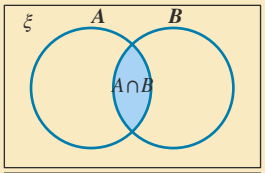
- The experimental probability is given by the following formula:

$$\text{Experimental Pr} = \frac{n(\text{successful trials})}{n(\text{trials})}$$

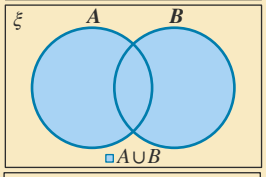
- The complement of an event A is denoted A' and is the set of outcomes that are not in A.
  - $n(A) + n(A') = n(\xi)$
  - $\Pr(A) + \Pr(A') = 1$

Intersection and union

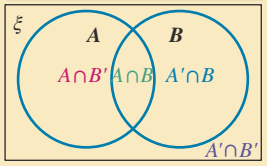
- The intersection of two events A and B is written $A \cap B$. These are the outcomes that are in A '**and**' in B.



- The union of two events A and B is written $A \cup B$. These are the outcomes that are in A '**or**' in B.



- A Venn diagram can be split into four distinct sections as shown at right.



- The Addition Law of probability states that:

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$$

Counting principles and factorial notation

- Counting techniques depend on whether order is important.
- The multiplication principle should be used when order is important.
- $n!$: the number of ways n distinct objects can be arranged
  $n! = n \times (n-1) \times (n-2) \times (n-3) \times \ldots \times 3 \times 2 \times 1$
- Permutations are used to count when order is important.

$${}^nP_r = \frac{n!}{(n-r)!}$$

- Combinations are used to count when order is *not* important.

$${}^nC_r = \frac{{}^nP_r}{r!}$$

Independent, dependent events and conditional probability

- Two events are **independent** if the outcome of one event does not affect the outcome of the other event.
- If A and B are independent, then:

$$\Pr(A \cap B) = \Pr(A) \times \Pr(B)$$

- Two events are **dependent** if the outcome of one event affects the outcome of the other event.
- Conditional probability applies to dependent events.
- The probability of A given that B has already occurred is given by:

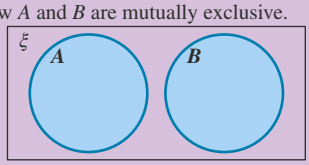
$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

- This can be rearranged into:

$$\Pr(A \cap B) = \Pr(A \mid B) \times \Pr(B)$$

Mutually exclusive events

- Mutually exclusive events have no elements in common.
- Venn diagrams show mutually exclusive events having no intersection. In the Venn diagram below A and B are mutually exclusive.

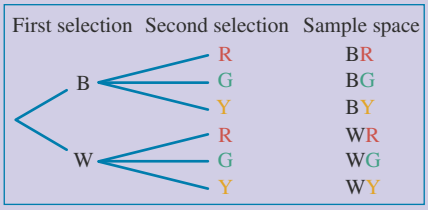


- For mutually exclusive events A and B:
  - $\Pr(A \cap B) = 0$
  - $\Pr(A \cup B) = \Pr(A) + \Pr(B)$

Multiple-step chance experiments

- Some chance experiments involve multiple trials performed separately.
- Two-way tables can be used to show the sample space of two-step experiments.
- Tree diagrams can be used to show the sample space and probabilities of multi-step experiments.

		Coin 1	
		H	T
Coin 2	H	HH	TH
	T	HT	TT



11.7.2 Success criteria

Tick the column to indicate that you have completed the lesson and how well you have understood it using the traffic light system.

(**Green:** I understand; **Yellow:** I can do it with help; **Red:** I do not understand)

Lesson	Success criteria			
11.2	I can use key probability terminology such as trials, frequency, sample space, and likely and unlikely events.			
	I can use two-way tables to represent sample spaces.			
	I can calculate experimental and theoretical probabilities.			
	I can determine the probability of complimentary events and mutually exclusive events.			
	I can use the addition rule to calculate the probability of event 'A or B'.			
11.3	I can create tree diagrams to represent two- and three-step chance experiments.			
	I can use tree diagrams to solve probability problems involving two or more trials or events.			
11.4	I can determine whether two events are dependent or independent.			
	I can calculate the probability of outcomes from two-step chance experiments involving dependent or independent events.			
11.5	I can determine from the language in a question whether it involves conditional probability.			
	I can use the rule for conditional probability to calculate other probabilities.			
11.6	I can apply the multiplication principle as a counting technique.			
	I can calculate permutations using factorials.			
	I can calculate permutations for situations involving like objects.			
	I can calculate the number of possible combinations in situations where order is not important.			

11.7.3 Project

Tricky dice

Dice games have been played throughout the world for many years. Professional gamblers resort to all types of devious measures in order to win. Often the other players are unaware of the tricks employed.



Imagine you are playing a game that involves rolling two dice. Instead of having each die marked with the numbers 1 to 6, let the first die have only the numbers 1, 2 and 3 (two of each) and the second die the numbers 4, 5 and 6 (two of each). If you were an observer to this game, you would see the numbers 1 to 6 occurring and probably not realise that the dice were not the regular type.

1. Complete the grid below to show the sample space on rolling these two dice.
2. Identify how many different outcomes there are. Compare this with the number of different outcomes using two regular dice.

Die 2 \ Die 1	1	2	3	1	2	3
4						
5						
6						
4						
5						
6						

3. Calculate the chance of rolling a double using these dice.
4. The numbers on the two dice are added after rolling. Complete the table below to show the totals possible.

Die 2 \ Die 1	1	2	3	1	2	3
4						
5						
6						
4						
5						
6						

5. Identify how many different totals are possible and list them.
6. State which total you have the greatest chance of rolling. State which total you have the least chance of rolling.
7. If you played a game in which you had to bet on rolling a total of less than 7, equal to 7 or greater than 7, explain which option would you be best to take.
8. If you had to bet on an even-number outcome or an odd-number outcome, explain which would be the better option.
9. The rules are changed to subtracting the numbers on the two dice instead of adding them. Complete the following table to show the outcomes possible.

Die 2 \ Die 1	1	2	3	1	2	3
4						
5						
6						
4						
5						
6						

10. Identify how many different outcomes are possible in this case and list them.
11. State the most frequently occurring outcome and how many times it occurs.
12. Devise a game of your own using these dice. On a separate sheet of paper, write out rules for your game and provide a solution, indicating the best options for winning.

Resources

eWorkbook Topic 11 Workbook (worksheets, code puzzle and project) (ewbk-2103)
Interactivities Crossword (int-9180)
Sudoku puzzle (int-3598)

Exercise 11.7 Review questions

learnon

Fluency

1. MC Choose which of the following is always true for an event, M, and its complementary event, M'.
   A. $\Pr(M) + \Pr(M') = 1$ **B.** $\Pr(M) - \Pr(M') = 1$
   C. $\Pr(M) + \Pr(M') = 0$ **D.** $\Pr(M) - \Pr(M') = 0$

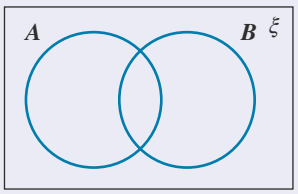
2. MC A number is chosen from the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Select which of the following pairs of events is mutually exclusive.
   A. $\{2, 4, 6\}$ and $\{4, 6, 7, 8\}$ **B.** $\{1, 2, 3, 5\}$ and $\{4, 6, 7, 8\}$
   C. $\{0, 1, 2, 3\}$ and $\{3, 4, 5, 6\}$ **D.** {multiples of 2} and {factors of 8}

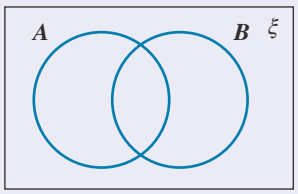
3. MC Choose which of the following states the Multiplication Law of probability correctly.
   A. $\Pr(A \cap B) = \Pr(A) + \Pr(B)$ **B.** $\Pr(A \cap B) = \Pr(A) \times \Pr(B)$
   C. $\Pr(A \cup B) = \Pr(A) \times \Pr(B)$ **D.** $\Pr(A \cup B) = \Pr(A) + \Pr(B)$

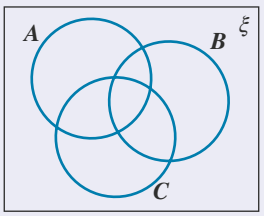
4. MC Given $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{2, 3, 4\}$ and $B = \{3, 4, 5, 8\}$, $A \cap B$ is:
   A. $\{2, 3, 3, 4, 4, 5, 8\}$ **B.** $\{3, 4\}$
   C. $\{2, 3, 4\}$ **D.** $\{2, 3, 4, 5, 8\}$

5. MC Given $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{2, 3, 4\}$ and $B = \{3, 4, 5, 8\}$, $A \cap B'$ is:
   A. $\{3, 4\}$ **B.** $\{2\}$ **C.** $\{2, 3, 4, 5, 8\}$ **D.** $\{2, 3, 4\}$

6. MC There are 12 people on the committee at the local football club. Determine how many ways can a president and a secretary be chosen from this committee.
   A. 2 **B.** 23 **C.** 132 **D.** 144

7. Shade the region stated for each of the following Venn diagrams.
   a. $A' \cup B$
   

   b. $A' \cap B'$
   

   c. $A' \cap B' \cap C$
   

8. **a.** State the definition of ${}^nP_r$.
   b. Without a calculator, compute the value of ${}^{10}P_3$.
   c. Prove that ${}^nP_r = \dfrac{n!}{(n-r)!}$.

Problem solving

9. MC From past experience, it is concluded that there is a 99% probability that July will be a wet month in Launceston (it has an average rainfall of approximately 80 mm). The probability that July will not be a wet month next year in Launceston is:

A. 99% **B.** 0.99 **C.** $\frac{1}{100}$ **D.** 1

10. MC A card is drawn from a well-shuffled deck of 52 cards. Select the theoretical probability of not selecting a red card.

A. $\frac{3}{4}$ **B.** $\frac{1}{4}$ **C.** $\frac{1}{13}$ **D.** $\frac{1}{2}$

11. MC Choose which of the following events is not equally likely.

A. Obtaining a 5 or obtaining a 1 when a die is rolled
B. Obtaining a club or obtaining a diamond when a card is drawn from a pack of cards
C. Obtaining 2 Heads or obtaining 2 Tails when a coin is tossed twice
D. Obtaining 2 Heads or obtaining 1 Head when a coin is tossed twice

12. MC The Australian cricket team has won 12 of the last 15 Test matches. Select the experimental probability of Australia not winning its next Test match.

A. $\frac{4}{5}$ **B.** $\frac{1}{5}$ **C.** $\frac{1}{4}$ **D.** $\frac{3}{4}$

13. A card is drawn from a well-shuffled pack of 52 cards. Calculate the theoretical probability of drawing:

a. an ace **b.** a spade **c.** a queen or a king **d.** not a heart.

14. A die is rolled five times.

a. Calculate the probability of rolling five 6s. **b.** Calculate the probability of not rolling five 6s.

15. Alan and Mary own 3 of the 8 dogs in a race. Evaluate the probability that:

a. one of Alan's or Mary's dogs will win **b.** none of Alan's or Mary's dogs will win.

16. A die is rolled. Event A is obtaining an even number. Event B is obtaining a 3.

a. Explain if events A and B are mutually exclusive.
b. Calculate $\Pr(A)$ and $\Pr(B)$.
c. Calculate $\Pr(A \cup B)$.

17. A card is drawn from a shuffled pack of 52 playing cards. Event A is drawing a club and event B is drawing an ace.

a. Explain if events A and B are mutually exclusive.
b. Calculate $\Pr(A)$, $\Pr(B)$ and $\Pr(A \cap B)$.
c. Calculate $\Pr(A \cup B)$.

18. A tetrahedral die is numbered 0, 1, 2 and 3. Two of these dice are rolled and the sum of the numbers (the number on the face that the die sits on) is taken.

a. Show the possible outcomes in a two-way table.
b. Determine if all the outcomes are equally likely.
c. Determine which total has the least chance of being rolled.
d. Determine which total has the best chance of being rolled.
e. Determine which sums have the same chance of being rolled.

19. A bag contains 20 pears, of which 5 are bad. Cathy picks 2 pears (without replacement) from the bag. Evaluate the probability that:
a. both pears are bad **b.** both pears are good **c.** one of the two pears is good.

20. Determine the probability of drawing 2 aces from a pack of cards if:
a. the first card is replaced before the second one is drawn
b. the first card drawn is not replaced.

21. A free-style snowboard competition has 15 entrants. Determine how many ways the first, second and third places can be filled. You may wish to use technology to answer this question.

22. The main cricket ground in Brisbane is called the Gabba, which is short for Woolloongabba. Determine how many different arrangements of letters can be made from the word WOOLLOONGABBA. You may wish to use technology to answer the question.

23. A committee of 5 men and 5 women is to be chosen from 8 men and 9 women. Determine how many ways this can be done.

Questions **24–26** refer to the following information.

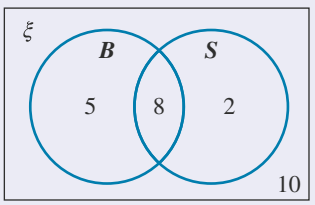
The Maryborough Tennis Championships involve 16 players. The organisers plan to use 3 courts and assume that each match will last on average 2 hours and that no more than 4 matches will be played on any court per day.

24. In a 'round robin' each player plays every other player once.
a. If the organisers use a round robin format, determine how many matches will be played in all.
b. Determine how long the tournament lasts.

25. The organisers split the 16 players into two pools of 8 players each. After a 'round robin' within each pool, a final is played between the winners of each pool.
a. Determine how many matches are played in the tournament.
b. Determine how long the tournament lasts.

26. A 'knock out' format is one in which the loser of every match drops out and the winners proceed to the next round until there is only one winner left.
a. If the game starts with 16 players, determine how many matches are needed before a winner is obtained.
b. Determine how long the tournament lasts.

27. On grandparents day at a school, a group of grandparents was asked where they most like to take their grandchildren — the beach (B) or shopping (S). The results are illustrated in the Venn diagram.



Use the Venn diagram to calculate the following probabilities relating to the place grandparents most like to take their grandchildren.
a. Determine the probability that a randomly selected grandparent preferred to take their grandchildren to the beach or shopping.
b. Determine the probability that a randomly selected grandparent liked to take their grandchildren to the beach, given that they liked to take their grandchildren shopping.

28. When all of Saphron's team players turn up for their twice weekly netball training, the chance that they then win their Saturday game is 0.65. If not all players are at the training session, then the chance of winning their Saturday game is 0.40.
Over a four-week period, Saphron's players all turn up for training three times.
a. Using a tree diagram, with T to represent all players training and W to represent a win, represent the winning chance of Saphron's netball team.
b. Using the tree diagram constructed in part **a**, determine the probability of Saphron's team winning their Saturday game.
Write your answer correct to 4 decimal places.
c. Determine the exact probability that Saphron's team did not train given that they won their Saturday game.

29. Andrew does not know the answer to two questions on a multiple-choice exam. The first question has four choices and the second question he does not know has five choices.
a. Determine the probability that he will get both questions wrong.
b. If he is certain that one of the choices cannot be the answer in the first question, determine how this will change the probability that he will get both questions wrong.

30. Mariah the Mathematics teacher wanted to give her students a chance to win a reward at the end of the term. She placed 20 cards into a box and wrote the word ON on 16 cards and OFF on 4 cards.
After a student chooses a card, that card is replaced into the box for the next student to draw. If a student chooses an OFF card, then they do not have to attend school on a specified day.
If they choose an ON card, then they do not receive a day off.
a. Mick, a student, chose a random card from the box. Calculate the probability he received a day off.
b. Juanita, a student, chose a random card from the box after Mick.
Calculate the probability that she did not receive a day off.
c. Determine the probability that Mick and Juanita both received a day off.

31. In the game of draw poker, a player is dealt 5 cards from a deck of 52. To obtain a flush, all 5 cards must be of the same suit.
a. Determine the probability of getting a diamond flush.
b. Determine the probability of getting any flush.

32. **a.** A Year 10 boy is talking with a Year 10 girl and asks her if she has any brothers or sisters. She says, 'Yes, I have one.'
Determine the probability that she has at least one sister.
b. A Year 10 boy is talking with a Year 10 girl and asks her if she has any brothers or sisters. She says, 'Yes, I have three.'
Determine the probability that she has at least one sister.



To test your understanding and knowledge of this topic, go to your learnON title at www.jacplus.com.au and complete the **post-test**.

Answers

Topic 11 Probability

11.1 Pre-test

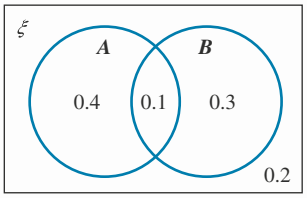
1. $\Pr(A \cap B) = 0.2$
2. $\Pr(A) = 0.51$
3. Independent
4. $\frac{1}{5}$
5. C
6. D
7. A
8. 0.56
9. A
10. False
11. $\Pr(B|A) = 0.2$
12. A
13. A
14. 0.031 25
15. A

11.2 Review of probability

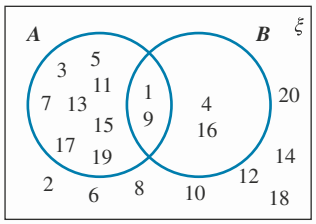
1. a. {I, II, III, IV, V, VI}
   b. Relative frequency for I = 0.2
   Relative frequency for II = 0.12
   Relative frequency for III = 0.16
   Relative frequency for IV = 0.14
   Relative frequency for V = 0.24
   Relative frequency for VI = 0.14
   c. The spinner should be spun a larger number of times.
2. a. 500 students
   b. Frequency for silver = 0.16
   c. Frequency for black and green = 0.316
   d. Blazing Blue
3. Sample responses can be found in the worked solutions in the online resources.
4. a.

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

   b. The sum of 7
   c. The sum of 2 or 12
5. $\frac{2}{3}$
6. a. $\frac{1}{13}$ b. $\frac{1}{4}$ c. $\frac{4}{13}$
7. Sample responses are given for part ii.
   a. i. No. There are many others foods one could have.
   ii. Having Weet Bix and not having Weet Bix
   b. i. No. There are other means of transport, for example catching a bus.
   ii. Walking to a friend's place and not walking to a friend's place
   c. i. No. There are other possible leisure activities.
   ii. Watching TV and not watching TV
   d. i. No. The number 5 can be rolled too.
   ii. Rolling a number less than 5 and rolling a number 5 or greater
   e. i. Yes. There are only two possible outcomes: passing or failing.
   ii. No change is required to make these events complementary.
8. a. $\frac{1}{13}$ b. $\frac{1}{4}$ c. $\frac{1}{2}$ d. $\frac{12}{13}$
   e. 0 f. $\frac{1}{2}$
9. a. $\frac{1}{5}$ b. $\frac{7}{20}$ c. $\frac{11}{20}$ d. 0
10. a. $\Pr(A \cap B) = 0.1$
    b.



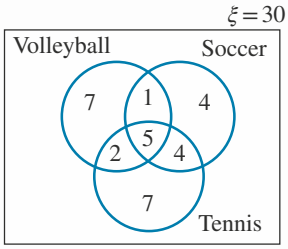
    c. $\Pr(A \cap B') = 0.4$
11. a. i. $\Pr(A \cup B) = 0.85$ ii. $\Pr(A \cap B)' = 0.95$
    b. C
12. a.



    b. i. $\frac{10}{20} = \frac{1}{2}$ ii. $\frac{4}{20} = \frac{1}{5}$ iii. $\frac{2}{20} = \frac{1}{10}$
    iv. $\frac{12}{20} = \frac{3}{5}$ v. $\frac{8}{20} = \frac{2}{5}$
13. a. No. For a 6-sided die, $\Pr(4) = \frac{1}{6}$; for an 8-sided die, $\Pr(4) = \frac{1}{8}$.
    b. Yes; $\Pr(\text{odd}) = \frac{1}{2}$ for both 6-sided and 8-sided dice
14. Yes; $\Pr(5) = \frac{1}{2}$, $\Pr(6) = \frac{1}{2}$.

15. Sample responses can be found in the worked solutions in the online resources.

16. a.



b. i. $\frac{1}{2}$ **ii.** $\frac{1}{6}$ **iii.** $\frac{1}{30}$ **iv.** $\frac{2}{5}$ **v.** $\frac{7}{15}$

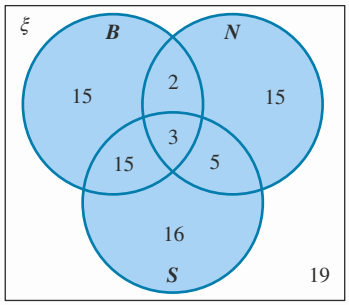
c. i. $\frac{1}{2}$ **ii.** $\frac{8}{15}$

17. Yes. Both have a probability of $\frac{1}{2}$.

18. Sample responses can be found in the worked solutions in the online resources.

19. 17 red and 1 purple, i.e. 18 more socks

20. a.



b. 19 students

c. 32 students

d. 15 students

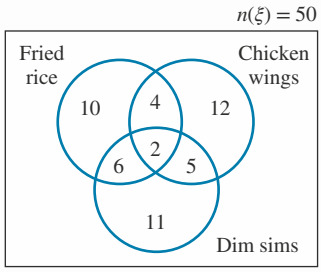
e. Frequency $= \frac{15}{90}$ or $\frac{1}{6}$

f. Probability $= \frac{3}{90} = \frac{1}{30}$

21. a. i. 50 **ii.** 7 **iii.** 25 **iv.** 8

b. i. $\frac{1}{2}$ **ii.** $\frac{3}{50}$ **iii.** $\frac{6}{25}$

c. i.



ii. $\frac{1}{25}$

22. a.

Die 2 outcomes

Die 1 outcomes	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

b. 6

c. No. The frequency of the numbers is different.

d.

Sum	2	3	4	5	6	7	8	9	10	11	12
Frequency	1	2	3	4	5	6	5	4	3	2	1

e. i. $\frac{1}{36}$ **ii.** $\frac{1}{6}$ **iii.** $\frac{1}{18}$

f. i. $\frac{1}{36}$ **ii.** $\frac{1}{6}$ **iii.** $\frac{1}{6}$

g. 50

23. Sample responses can be found in the worked solutions in the online resources.

11.3 Tree diagrams

1. If the probabilities of two events are different, the first column of branches indicates the probabilities for the first event and the second column of branches indicates the probabilities for the second event. The product of each branch gives the probability. All probabilities add to 1.

2. a. 12 different outcomes

b. No. Each branch is a product of different probabilities.

c. Less likely

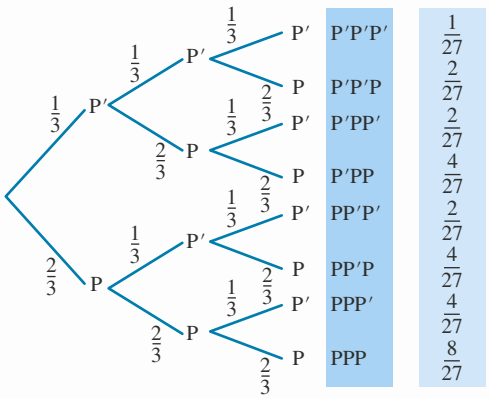
d. Indigo elephant

e. i. Pr(Blue elephant) $= \frac{7}{40}$

ii. Pr(Indigo elephant) $= \frac{7}{20}$

iii. Pr(Donkey) $= \frac{1}{5}$

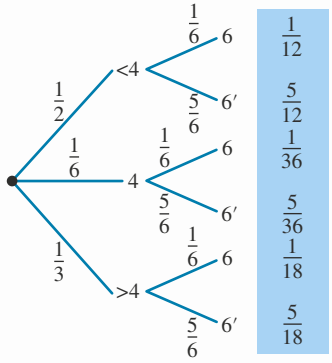
3. a.



b. i. $\frac{8}{27}$ **ii.** $\frac{4}{27}$ **iii.** $\frac{12}{27}$

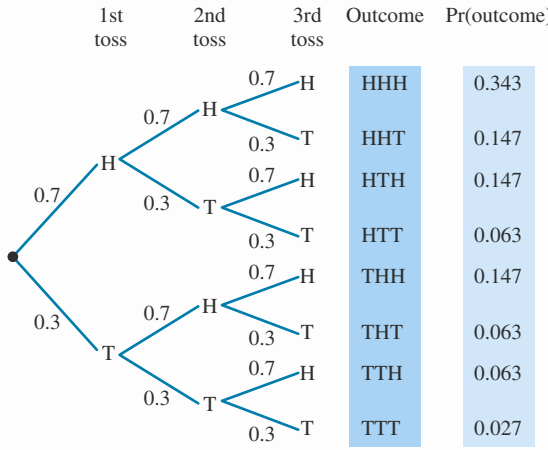
4. a. i. $\frac{1}{2}$ **ii.** $\frac{1}{6}$ **iii.** $\frac{5}{6}$

b.



c. i. $\frac{1}{36}$ **ii.** $\frac{1}{12}$ **iii.** $\frac{5}{36}$ **iv.** $\frac{5}{18}$

5. a.

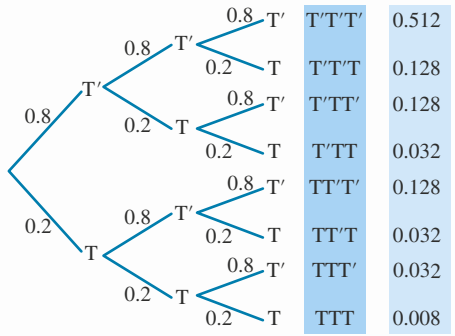


b. Pr(HHH) = 0.343

c. Pr(at least 1 Tail) = 0.657

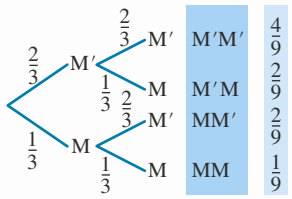
d. Pr(exactly 2 Tails) = 0.189

6. a.



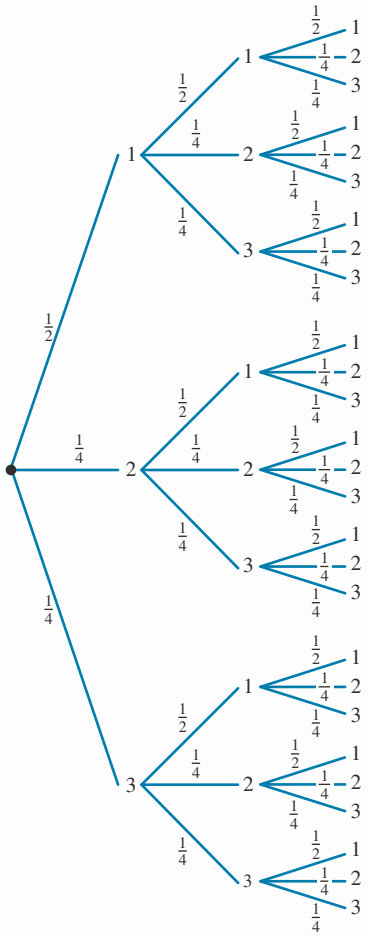
b. 0.520

7. a.



b. $\frac{1}{9}$

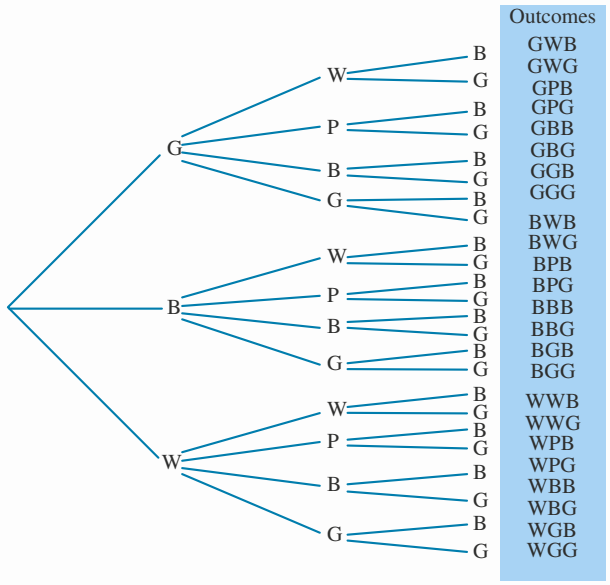
8. a.



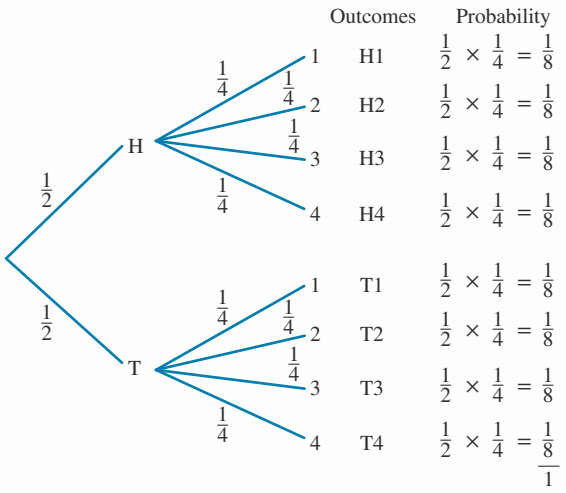
b. $\frac{3}{8}$

c. $\frac{7}{8}$

9. a.


G
B
W
W
P
B
G
B
G
B
G

b.



c. $\frac{1}{4}$

14. a. $\frac{256}{625}$ **b.** $\frac{16}{625}$

c. $\frac{96}{256}$ **d.** $\frac{17}{625}$

11.4 Independent and dependent events

1. a. 0.28 **b.** 0.12 **c.** 0.42 **d.** 0.18

2. Dependent

3. Independent

4. a. Yes, the outcomes are independent.

b. i. $\frac{1}{2}$ **ii.** $\frac{1}{6}$

c. $\frac{1}{12}$

5. $\frac{1}{40}$

6. $\frac{5}{36}$

7. a. $\frac{16}{25}$ **b.** $\frac{64}{125}$

c. $\frac{1}{25}$ **d.** $\frac{4}{25}$

8. a. C **b.** D

9. a. $\frac{3}{77}$ **b.** $\frac{48}{77}$

c. $\frac{8}{77}$ **d.** $\frac{18}{77}$

10. a. $\frac{1}{37}$ **b.** $\frac{1}{1369}$ **c.** $\frac{73}{1369}$

11. 0.9

12. $\frac{1}{14}$

13. a. $\frac{1}{5}$ **b.** $\frac{1}{5}$

c. $\frac{1}{10}$ **d.** $\frac{1}{3}$

14. a. $\frac{1}{17}$ **b.** $\frac{1}{221}$ **c.** $\frac{25}{102}$

15. a. $\frac{26}{145}$ **b.** $\frac{136}{435}$ **c.** $\frac{221}{435}$

16. No. Coin tosses are independent events. No one toss affects the outcome of the next. The probability of a Head or Tail on a fair coin is always 0.5. Greg has a 50% chance of tossing a Head on the next coin toss as was the chance in each of the previous 9 tosses.

17. No. As events are illustrated on a tree diagram, the individual probability of each outcome is recorded. The probability of a dependent event is calculated (altered according to the previous event) and can be considered as if it was an independent event. As such, the Multiplication Law of probability can be applied along the branches to calculate the probability of successive events.

18. Only if $\Pr(A) = 0$ or $\Pr(B) = 0$ can two events be independent and mutually exclusive. For an event to have a probability of 0 means that it is impossible, so it is a trivial scenario.

19. a. A and B are independent.

b. A and B are not independent in this situation.

20. $\frac{5}{17}$

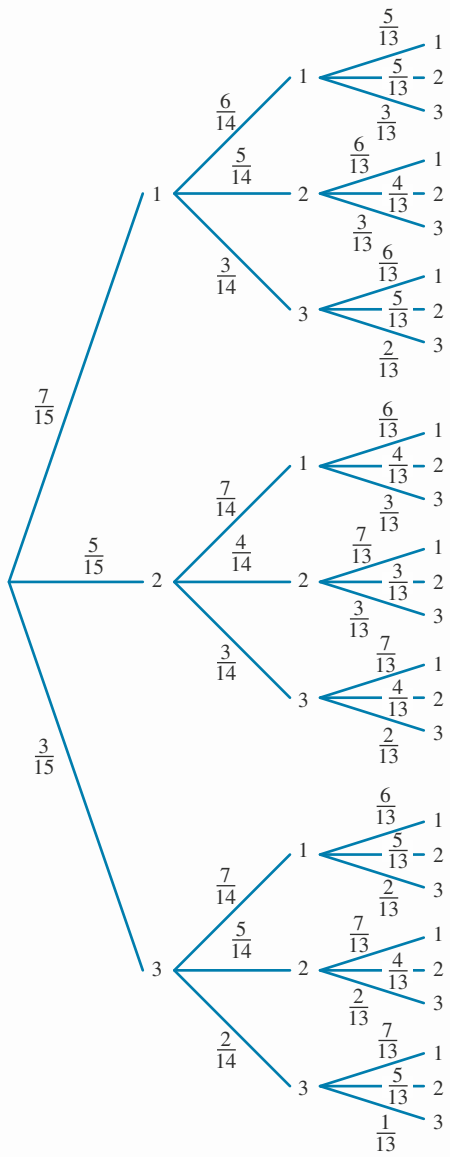
21. a. Dependent

b. $\Pr(1) = \frac{7}{15}$

$\Pr(2) = \frac{5}{15} = \frac{1}{3}$

$\Pr(3) = \frac{3}{15} = \frac{1}{5}$

c.



d. i. $\Pr(1, 1, 1) = \frac{1}{13}$ **ii.** $\Pr(2, 2, 2) = \frac{2}{91}$

iii. $\Pr(3, 3, 3) = \frac{1}{455}$ **iv.** $\Pr(\text{at least one 3}) = \frac{47}{91}$

22. a. $\Pr(9) = \frac{1}{455}$

$\Pr(7 \text{ or } 8) = \frac{66}{455}$

$\Pr(5 \text{ or } 6) = \frac{248}{455}$

$\Pr(3 \text{ or } 4) = \frac{4}{13}$

b. \$393.41

11.5 Conditional probability

1. a. $\Pr(\text{J}) = \frac{41}{90}$ **b.** $\Pr(z|\text{J}) = \frac{12}{41}$

2. a. $\Pr(\text{S}) = \frac{13}{30}$ **b.** $\Pr(\text{S}|(\text{C} \cup \text{S})) = \frac{13}{28}$

3. a. 0.3 **b.** $\frac{3}{7}$

4. a. $\frac{9}{13}$ **b.** $\frac{3}{5}$

5. 0.58 or $\frac{15}{26}$

6. 0.22 or $\frac{5}{23}$

7. 0.9

8. 0.2051

9. A

10. a. D

b. i. $\Pr(A|B) = \frac{1}{6}$ **ii.** $\Pr(B|A) = \frac{1}{6}$

iii. $\Pr(C|A) = \frac{1}{6}$ **iv.** $\Pr(C|B) = 0$

11. a. $\frac{109}{232}$ **b.** $\frac{73}{116}$ **c.** $\frac{23}{86}$ **d.** $\frac{22}{45}$

12. Conditional probability is when the probability of one event depends on the outcome of another event.

13. a. 0.0875

b. 0.065

c. i. 0.585 **ii.** 15.25%

14. a. i. $\frac{1}{2}$ **ii.** $\frac{2}{5}$ **iii.** $\frac{1}{5}$ **iv.** $\frac{1}{2}$ **v.** $\frac{2}{5}$

b. Yes, A and B are independent.

c. If $\Pr(A|B) = \frac{\Pr(A \cap B)}{\Pr(B)} = \Pr(A)$, then A and B will be independent events. This is because the probability of A given B occurs is the same as the probability of A, meaning the probability of A is independent of B occurring.

15. 8.7% chance

16. 0.847

17. 0.61

11.6 Counting principles and factorial notation (10A)

1. a. AB AC
BA BC
CA CB

b. 6

c. $\frac{1}{3}$

d. 9

2. BG BY BR
GB GY GR
YB YG YR
RB RG RY

3. ABC ACB BAC
BCA CAB CBA

4. **a.** 42 **b.** 210 **c.** 840 **d.** 2520 **e.** 360

5. **a.** 8 **b.** 32 **c.** 512

6. **a.** $4! = 4 \times 3 \times 2 \times 1$

b. $5! = 5 \times 4 \times 3 \times 2 \times 1$

c. $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1$

d. $7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$

7. **a.** 3024

b. 151 200

c. 840

d. 720

8. **a.** 3 628 800

b. 87 178 291 200

c. 362 880

d. 20 922 789 888 000

9. **a.** ${}^8P_2 = 8 \times 7 = 56$

b. ${}^7P_5 = 7 \times 6 \times 5 \times 4 \times 3 = 2520$

c. ${}^8P_7 = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 = 40\,320$

10. **a.** ${}^9P_6 = \dfrac{9!}{(9-6)!}$

$= \dfrac{9!}{3!}$

$= \dfrac{3\,62\,880}{6}$

$= 60\,480$

b. ${}^5P_2 = \dfrac{5!}{(5-2)!}$

$= \dfrac{5!}{3!}$

$= \dfrac{120}{6}$

$= 20$

c. ${}^{18}P_5 = \dfrac{18!}{(18-5)!}$

$= \dfrac{18!}{13!}$

$= \dfrac{6\,402\,373\,705\,728\,000}{6\,227\,020\,800}$

$= 1\,028\,160$

11. **a.** 27 907 200 **b.** 639 200 **c.** 1 028 160

12. 10

13. 1260

14. **a.** ${}^8C_3 = \dfrac{{}^8P_3}{3!}$ **b.** ${}^{19}C_2 = \dfrac{{}^{19}P_2}{2!}$

c. ${}^1C_1 = \dfrac{{}^1P_1}{1!}$ **d.** ${}^5C_0 = \dfrac{{}^5P_0}{0!}$

15. **a.** 24 **b.** 6

c. 12 **d.** 24

16. **a.** 200 **b.** 40 **c.** 50

17. **a.** 56

b. $\dfrac{1}{8}$

18. 83 160

19. 27 720

20. **a.** 1 **b.** 20 **c.** 120 **d.** 220

21. 100

22. 6

23. 256

24. 358 800

25. 120

26. 3024

27. 1 307 504

28. **a.** 2002 **b.** 840

29. **a.** **i.** ${}^{12}C_3 = 220, {}^{12}C_9 = 220$

ii. ${}^{15}C_8 = 6435, {}^{15}C_7 = 6435$

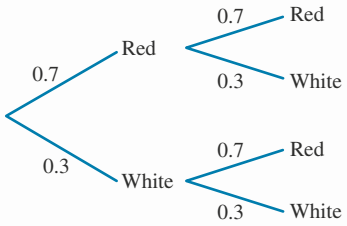
iii. ${}^{10}C_1 = 10, {}^{10}C_9 = 10$

iv. ${}^8C_3 = 56, {}^8C_5 = 56$

b. The value of ${}^nC_r$ is the same as ${}^nC_{n-r}$.

30. **a.** 8 145 060 **b.** 11 480

31. **a.**



b. 10 balls

c. 10 balls

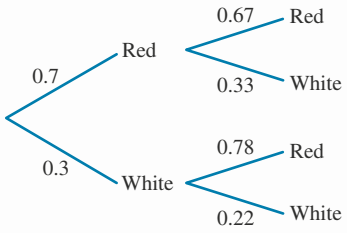
d. No; the events are independent.

e. P(RR) = 0.49

f. 9 balls

g. Yes. One ball has been removed from the bag.

h.



i. $P(RR) = \dfrac{7}{15} = 0.46\dot{}$

32. **a.** 1000 **b.** 27

c.

271	371	471
272	372	472
273	373	473
281	381	481
282	382	482
283	383	483
291	391	491
292	392	492
293	393	493

Project

1.

Die 2 \ Die 1	1	2	3	1	2	3
4	(1, 4)	(2, 4)	(3, 4)	(1, 4)	(2, 4)	(3, 4)
5	(1, 5)	(2, 5)	(3, 5)	(1, 5)	(2, 5)	(3, 5)
6	(1, 6)	(2, 6)	(3, 6)	(1, 6)	(2, 6)	(3, 6)
4	(1, 4)	(2, 4)	(3, 4)	(1, 4)	(2, 4)	(3, 4)
5	(1, 5)	(2, 5)	(3, 5)	(1, 5)	(2, 5)	(3, 5)
6	(1, 6)	(2, 6)	(3, 6)	(1, 6)	(2, 6)	(3, 6)

2. 9

3. 0

4.

Die 2 \ Die 1	1	2	3	1	2	3
4	5	6	7	5	6	7
5	6	7	8	6	7	8
6	7	8	9	7	8	9
4	5	6	7	5	6	7
5	6	7	8	6	7	8
6	7	8	9	7	8	9

5. 5; 5, 6, 7, 8, 9

6. 7; 5, 9

7. Equal to 7; probability is the highest.

8. Odd-number outcome; probability is higher.

9.

Die 2 \ Die 1	1	2	3	1	2	3
4	3	2	1	3	2	1
5	4	3	2	4	3	2
6	5	4	3	5	4	3
4	3	2	1	3	2	1
5	4	3	2	4	3	2
6	5	4	3	5	4	3

10. 5; 1, 2, 3, 4, 5,

11. 3; 12

12. Students need to apply the knowledge of probability and create a new game using dice given. They also need to provide rules for the game and solution, indicating the best options for winning.

11.7 Review questions

1. A

2. B

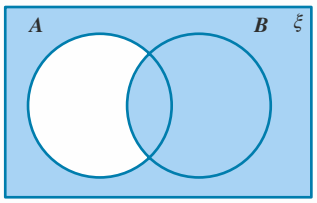
3. B

4. B

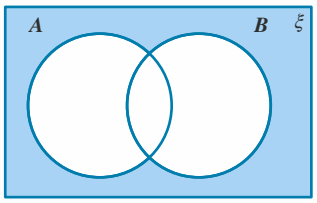
5. B

6. C

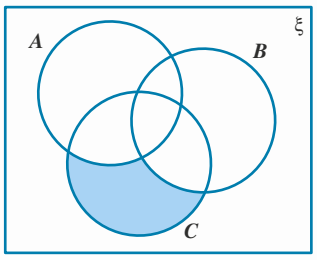
7. a.



b.



c.



8. a. ${}^{n}\mathrm{P}_{r}$ is the number of ways of choosing r objects from n distinct things when order is important. ${}^{n}\mathrm{P}_{r} = \frac{n!}{(n-r)!}$

b. ${}^{10}\mathrm{P}_{3} = 720$

c. Sample responses can be found in the worked solutions in the online resources.

9. C

10. D

11. D

12. B

13. a. $\frac{1}{13}$ **b.** $\frac{1}{4}$

c. $\frac{2}{13}$ **d.** $\frac{3}{4}$

14. a. $\frac{1}{7776}$

b. $\frac{7775}{7776}$

15. a. $\frac{3}{8}$

b. $\frac{5}{8}$

16. a. Yes. It is not possible to roll an even number and for that number to be a 3.

b. $\Pr(A) = \frac{1}{2}$ and $\Pr(B) = \frac{1}{6}$

c. $\frac{2}{3}$

17. a. No. It is possible to draw a card that is a club and an ace.

b. $\Pr(A) = \frac{1}{4}$ and $\Pr(B) = \frac{1}{13}$, $\Pr(A \cap B) = \frac{1}{52}$

c. $\frac{4}{13}$

18. a.

Die 1 outcomes \ Die 2 outcomes	**0**	**1**	**2**	**3**
0	0	1	2	3
1	1	2	3	4
2	2	3	4	5
3	3	4	5	6

b. No

c. 0 and 6

d. 3

e. 0 and 6, 1 and 5, 2 and 4

19. a. $\frac{1}{19}$

b. $\frac{21}{38}$

c. $\frac{15}{38}$

20. a. $\frac{1}{169}$

b. $\frac{1}{221}$

21. 2730

22. 32 432 400

23. 7056

24. a. 120 **b.** 10 days

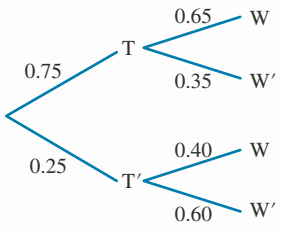
25. a. 57 **b.** 4 days 6 hours

26. a. 15 **b.** 1 day 4 hours

27. a. $\frac{15}{25} = \frac{3}{5}$

b. $\frac{8}{10} = \frac{4}{5}$

28. a.



b. 0.5875

c. $\frac{8}{47}$

29. a. $\frac{3}{5}$

b. $\frac{8}{15}$

30. a. $\frac{1}{5}$

b. $\frac{4}{5}$

c. $\frac{1}{25}$

31. a. 0.000 495

b. 0.001 981

32. a. $\frac{1}{2}$

b. $\frac{7}{8}$